*remote sensing*

*Review*

# Remote Sensing and Geospatial Analysis in the Big Data Era: A Survey

**Elias Dritsas and Maria Trigka ***

Industrial Systems Institute (ISI), Athena Research and Innovation Center, 26504 Patras, Greece; dritsas@isi.gr
* Correspondence: trigka@isi.gr

**Abstract:** The present survey examines the role of big data analytics in advancing remote sensing and geospatial analysis. The increasing volume and complexity of geospatial data are driving the adoption of machine learning (ML) and artificial intelligence (AI) techniques, such as convolutional neural networks (CNNs) and long short-term memory (LSTM) networks, to extract meaningful insights from large, diverse datasets. These AI methods enhance the accuracy and efficiency of spatial and temporal data analysis, benefiting applications in environmental monitoring, urban planning, and disaster management. Despite these advancements, challenges related to computational efficiency, data integration, and model transparency remain. This paper also discusses emerging trends and highlights the potential of hybrid approaches, cloud computing, and edge processing in overcoming these challenges. The integration of AI with geospatial data is poised to significantly improve our ability to monitor and manage Earth systems, supporting more informed and sustainable decision-making.

**Keywords:** remote sensing; geospatial analysis; big data; artificial intelligence; machine learning

check for updates

Academic Editors: Robert Tenzer and Hok Sum Fok

Received: 15 December 2024
Revised: 29 January 2025
Accepted: 3 February 2025
Published: 6 February 2025

**Citation:** Dritsas, E.; Trigka, M. Remote Sensing and Geospatial Analysis in the Big Data Era: A Survey. *Remote Sens.* **2025**, *17*, 550. https://doi.org/10.3390/rs17030550



## 1. Introduction

Remote sensing, as a field, has undergone significant advancements since its inception, evolving from basic aerial photography to the modern acquisition of high-resolution multispectral and hyperspectral data from sophisticated satellite constellations, unmanned aerial vehicles (UAVs), and ground-based sensors. Traditionally, remote sensing has provided critical insights into the earth's physical, biological, and chemical states, with applications spanning agriculture, forestry, climate monitoring, urban planning, and resource management. The primary objective of remote sensing has remained the same: to observe, quantify, and analyze the earth's surface and atmospheric conditions with the highest accuracy and efficiency [1,2].

However, the landscape of remote sensing has shifted dramatically in recent years. The confluence of rapid advancements in sensor technologies, decreasing costs of satellite deployments, and the widespread adoption of Internet of Things (IoT) networks have created an influx of diverse and voluminous geospatial data. This shift has transformed the field, moving it into what is widely known as the "big data era" in geospatial sciences. In contrast to traditional datasets, big geospatial data are characterized not only by their sheer size but also by their high frequency (temporal resolution), heterogeneity (data types and formats), and high dimensionality (multispectral or hyperspectral channels). Each characteristic presents distinct challenges regarding data management, processing, and analysis [3,4].

The rise of big data in remote sensing has also catalyzed the integration of advanced computational techniques, such as AI and ML, that can handle the scale and complexity of modern geospatial data. ML models—particularly deep learning (DL) architectures such as CNNs for image analysis and recurrent neural networks (RNNs) for temporal data—have introduced new capabilities in extracting meaningful insights from vast, unstructured datasets. These models can process and interpret complex spatial–temporal patterns with a degree of accuracy and granularity that would be unfeasible through traditional analytical methods [5,6].

The big data era has not only expanded the capabilities of remote sensing but has also democratized access to powerful computational resources. Cloud computing platforms like Google Earth Engine, Amazon Web Services, and Microsoft Azure provide scalable infrastructure that allows researchers and practitioners to store, process, and analyze petabytes of geospatial data without requiring extensive local resources. This development has facilitated broader access to data and computational tools, enabling various applications in resource-constrained contexts, such as real-time disaster management and precision agriculture in remote regions [7–9].

Despite the opportunities offered by big data in remote sensing, these developments are not without significant challenges. Managing, storing, and processing such massive and complex datasets require robust computational architectures, efficient data pipelines, and interoperability standards to ensure effective data integration across platforms. Data privacy and security issues, especially with the growing use of UAVs and ground-based sensors in urban areas, have introduced ethical and regulatory considerations that further complicate data governance. Moreover, the increasing reliance on AI models raises concerns regarding model interpretability, as complex neural networks often operate as "black boxes", limiting transparency and potentially leading to biased or erroneous outputs [10–12].

The motivation for this survey stems from the rapid increase in remote sensing data volume, driven by advancements in satellite, UAV, and IoT technologies. Managing and analyzing these data requires scalable solutions to handle their size, heterogeneity, and real-time demands. Big data analytics and AI have shown promise in extracting valuable insights but present technical and ethical challenges. This article aims to bridge gaps in understanding modern data processing methods, ML applications, and data fusion in remote sensing. Ultimately, it seeks to guide future research in addressing these complexities and leveraging new technologies for improved geospatial analysis. More specifically, the main contributions of this survey are the following:

- Provides a comprehensive overview of geospatial data sources in the era of big data, including satellite, UAV, and IoT data, and their unique characteristics.
- Examines advanced data processing and AI techniques, such as ML and DL, for enhancing remote sensing applications in various domains.
- Highlights transformative applications in environmental monitoring, disaster management, urban planning, and precision agriculture, showcasing how big data analytics is transforming these fields. Also, it notes case studies and application domains.
- Discusses key challenges in remote sensing and geospatial analysis, such as data management, computational efficiency, and the need for model interpretability, along with proposed solutions.
- Notes future research directions in geospatial science, emphasizing the integration of quantum computing, federated learning (FL), and advanced data fusion techniques to overcome current limitations.

The remainder of the paper is illustrated in Figure 1 and is structured as follows. Section 2 provides an overview of the primary data sources in remote sensing, focusing on data characteristics critical in a big data context. Moreover, Section 3 details the processing architectures, ML applications, and data integration techniques essential for modern geospatial analysis. Section 4 reviews real-world applications across diverse domains. Section 5 notes case studies and application domains. Furthermore, Section 6 discusses current challenges and forecasts future research directions. Finally, Section 7 concludes the survey.

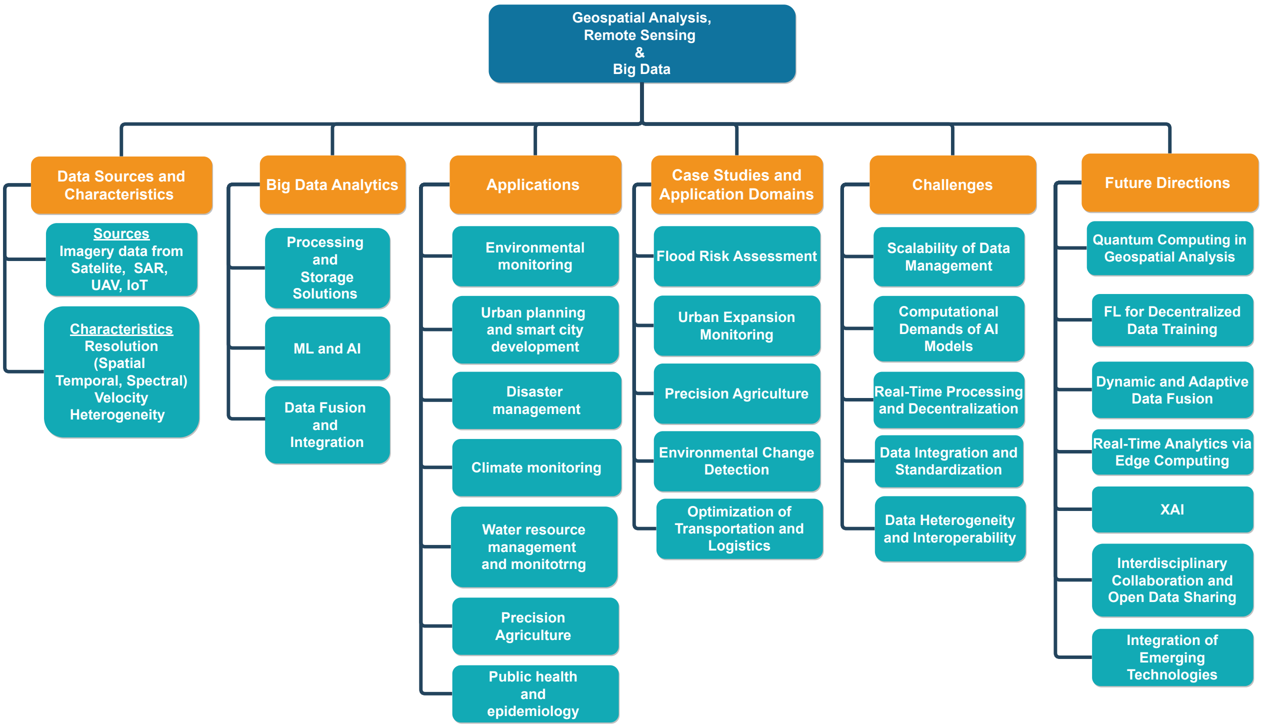


**Figure 1.** An overview of surveyed key topics for geospatial data analysis and remote sensing in the era of big data.

## 2. Remote Sensing and Geospatial Data Sources in the Big Data Era

The exponential growth in remote sensing and geospatial data sources marks a new era in big data, bringing unprecedented opportunities and challenges for environmental monitoring, urban planning, and more. These data sources, encompassing satellites, UAVs, and IoT networks, contribute vast amounts of spatial, temporal, and spectral information, driving the need for sophisticated processing and integration techniques in geospatial analysis.

### 2.1. Overview of Data Sources

The field of remote sensing has seen a dramatic expansion in data sources over recent years, contributing to a diverse and complex landscape of geospatial information. Satellite imagery remains a foundational pillar, providing comprehensive, multi-scale data that enable continuous monitoring of the Earth's surface. Advances in sensor technology have resulted in satellites equipped with high-resolution optical, multispectral, and hyperspectral sensors, offering unprecedented detail and enabling applications across environmental monitoring, agriculture, and urban planning. Additionally, synthetic aperture radar (SAR) technology, which captures imagery regardless of weather or lighting conditions, has expanded the versatility of satellite data by providing reliable insights even in challenging environments [13–15].

Alongside satellites, UAVs have become essential in remote sensing, offering high-resolution imagery and flexible data acquisition capabilities at customizable altitudes and intervals. UAVs are particularly valuable in areas where fine spatial resolution is required or where access is limited, such as precision agriculture, infrastructure inspection, and disaster response. The ability to capture images at user-defined intervals allows UAVs to fill temporal gaps between satellite passes, providing near-continuous monitoring in critical applications. Furthermore, UAVs can be equipped with various sensors, including RGB, multispectral, and thermal cameras, allowing for tailored data collection based on specific application requirements [16–18].

IoT has also made significant contributions to the remote sensing ecosystem through the deployment of ground-based sensors. IoT networks collect hyperlocal, high-frequency data on environmental parameters such as temperature, humidity, air quality, and soil moisture. These sensors add a new dimension to geospatial data by providing ground-truth measurements that complement and validate satellite and UAV observations. When integrated with remote sensing data, IoT sensor networks enhance data granularity and real-time applicability, particularly in dynamic environments such as urban areas and agricultural fields. The growth of IoT devices has fostered a shift towards continuous, high-resolution monitoring, enabling near-real-time insights and decision-making capabilities in critical sectors [19–21].

The diversity of remote sensing data sources contributes to a vast, multifaceted dataset ecosystem. Each source—satellites, UAVs, and IoT sensors—brings distinct advantages and challenges, contributing to an environment rich in spatial, temporal, and spectral information. However, this abundance of data is accompanied by complexities in data management, integration, and analysis. Satellite data offer broad coverage but vary in temporal resolution, with revisit times ranging from days to weeks depending on the satellite's orbit and configuration. UAVs provide high spatial resolution but are limited by battery life, regulatory constraints, and operational costs, which can restrict their use to specific locations or periods. IoT sensors, while providing continuous data at the local level, require extensive network infrastructure and can be limited in spatial coverage, depending on the density and distribution of deployed sensors [22–24].

In sum, the integration of diverse data sources is foundational to the advancement of remote sensing in the big data era. By leveraging satellite imagery, UAVs, and IoT sensors, researchers and practitioners can capture a holistic view of dynamic landscapes, providing insights that are spatially expansive and temporally precise. The convergence of these data sources not only enriches our understanding of complex environmental and societal systems but also underscores the need for sophisticated data processing and fusion techniques to harness the full potential of big data in geospatial science.

### *2.2. Data Characteristics*

Remote sensing data are characterized by a rich combination of spatial, temporal, and spectral dimensions, each contributing to the complexity and analytical potential of geospatial information in the big data era. Spatial resolution, which defines the level of detail in an image, varies widely across data sources and is crucial for accurately observing features at different scales. High-resolution satellite imagery and UAV data allow for the detection of fine details, such as individual trees or small buildings, which are essential for applications like urban planning, precision agriculture, and environmental monitoring. In contrast, coarser spatial resolutions, often associated with large-scale satellite missions, offer broad coverage suitable for regional or global analysis, making them invaluable for tracking large-scale phenomena such as deforestation, climate patterns, and urban sprawl [25–27].

Temporal resolution is another critical characteristic of remote sensing data, referring to the frequency at which data are acquired over a specific location. The big data era has brought improvements in temporal resolution, particularly with the advent of satellite constellations and UAV technologies capable of capturing data at user-defined intervals. High temporal resolution is indispensable for monitoring rapidly changing events, such as natural disasters, crop growth cycles, and traffic patterns. However, differences in temporal resolution across data sources can present integration challenges; while ground-based IoT sensors may provide continuous real-time data, satellite revisit intervals vary, potentially leaving gaps in time-sensitive applications. These variations necessitate data fusion and interpolation techniques to create seamless temporal datasets, which are vital for real-time monitoring and decision making [28–30].

Spectral resolution, which denotes the range and specificity of wavelengths captured, adds yet another dimension of complexity. Remote sensing data span a spectrum from visible light to infrared and microwave, with each wavelength band offering distinct insights. Multispectral and hyperspectral imaging, common in advanced satellite and UAV systems, captures information across dozens or even hundreds of spectral bands, allowing for detailed analysis of vegetation health, mineral compositions, water quality, and more. Hyperspectral data, in particular, provide a unique spectral signature for each material, facilitating precise identification of surface materials and conditions. However, the high dimensionality of hyperspectral data requires specialized processing techniques to manage storage, reduce noise, and interpret spectral information effectively [31–33].

The data generated by remote sensing are not only vast in volume but also highly heterogeneous, comprising various formats, resolutions, and data structures. This heterogeneity reflects the diversity of sensors, each with distinct characteristics tailored to specific applications. Integrating data from multiple sources poses significant challenges, especially when considering the need for standardization across data formats and metadata structures. This complexity extends to issues of data quality, with each data source introducing its own potential for noise, distortions, and inconsistencies, further complicating data processing and interpretation [34–36].

Another defining characteristic is the velocity of data acquisition, particularly in real-time or near-real-time applications. With the rapid deployment of IoT sensors and the development of edge computing capabilities, data can now be collected and processed at the source, allowing for immediate insights in dynamic environments. This is especially valuable in applications such as disaster response, where rapid access to data can aid in timely resource allocation and damage assessment. However, managing and processing high-velocity data streams requires robust computational architectures that can handle both the speed and the volume of incoming data without compromising accuracy [37–39].

In essence, the characteristics of remote sensing data in the big data era present both opportunities and challenges. These datasets' diverse spatial, temporal, and spectral resolutions enable a multidimensional understanding of the world, yet they demand advanced processing techniques and infrastructure to harness their full potential. As geospatial analysis continues to expand in scope and scale, developing methods that can efficiently handle the intricate characteristics of remote sensing data will be essential for advancing applications in environmental science, urban planning, agriculture, and beyond.

As discussed, the exponential growth in remote sensing data sources and their diverse characteristics has ushered in unprecedented opportunities and challenges for geospatial analysis. To provide a consolidated view of the studies and references underpinning this discussion, Table 1 offers a detailed summary of key works. It highlights significant advancements in satellite, UAV, and IoT data sources, as well as methodologies for addressing spatial, temporal, and spectral complexities in the big data era. Finally, Table 2 provides a comparative summary of key remote sensing data sources, highlighting their spatial and temporal characteristics, primary applications, and associated challenges in geospatial analysis.

**Table 1.** Key references highlighting data sources and characteristics for remote sensing in the big data era.

| Reference | Description | Data Modality | Processing Requirements |
|---|---|---|---|
| [13–15] | Discuss geospatial interpretation, fundamental principles, and advancements in satellite technology for environmental and urban monitoring. | Satellite imagery, multispectral, SAR | Cloud processing for large-scale datasets, AI frameworks |
| [16–18] | Focus on applications like bridge condition assessment, semantic image descriptions, and flexible UAV data collection platforms. | UAV imagery, thermal, multispectral | Edge computing for UAV data analysis |
| [19–21] | Cover hyperlocal temperature mapping, IoT sensor deployment, and distributed frameworks for data integration. | IoT sensor data (temperature, air quality, soil moisture) | Real-time IoT integration, decentralized processing |
| [22–24] | Explore challenges and techniques in integrating multi-source temporal data and enhancing resolution through fusion. | Multisource temporal datasets, fused resolutions | High-performance computing for data fusion |
| [25–27] | Highlight datasets and tools for scene understanding and spatial analysis using high-resolution imagery. | High-resolution optical imagery | Graphics processing units (GPUs)/ML tools for image analysis |
| [28–30] | Examine the role of remote sensing in tracking species diversity, ecological processes, and habitat conservation. | Satellite imagery for biodiversity tracking | Temporal analysis frameworks, biodiversity models |
| [31–33] | Discuss water quality monitoring systems and the use of edge computing for real-time environmental assessments. | Hyperspectral water quality monitoring, edge data | Edge–cloud hybrid systems |
| [34–36] | Address dynamic spatial–temporal processing, real-time data streaming, and AI-driven geospatial insights. | Dynamic multi-temporal streams | Streaming platforms, AI pipelines |
| [37–39] | Investigate AI models and frameworks for handling big data in earth observation (EO) and remote sensing. | AI-driven fusion of multi-sensor data | Distributed AI frameworks for large-scale analysis |

**Table 2.** Comparative characteristics of remote sensing data sources.

| Data Source | Spatial Resolution | Temporal Resolution | Primary Applications | Key Challenges |
|---|---|---|---|---|
| Satellite Imagery | Moderate to high (e.g., 10–30 m for Landsat, sub-meter for commercial satellites) | Days to weeks (varies by orbit and satellite constellation) | Environmental monitoring, urban growth analysis, precision agriculture, and disaster management | Weather dependency (optical sensors), limited revisit frequency, high costs for high-resolution data |
| UAVs (Drones) | Very high (centimeter level) | On-demand (as needed) | Precision agriculture, infrastructure inspection, real-time disaster response | Restricted flight duration (battery life), limited coverage, and regulatory constraints |

**Table 2.** *Cont.*

| Data Source | Spatial Resolution | Temporal Resolution | Primary Applications | Key Challenges |
| --- | --- | --- | --- | --- |
| IoT Sensors | Point measurements (localized data) | Real-time or near-real-time | Smart cities (e.g., traffic and pollution monitoring), agricultural soil and water management | Limited spatial coverage, data heterogeneity, and reliance on network connectivity |
| SAR | Moderate to high (depends on platform and system design) | All-weather, day-and-night imaging | Flood mapping, deforestation detection, urban infrastructure monitoring | Computationally intensive data processing and specialized expertise for interpretation |

## 3. Big Data Analytics in Remote Sensing and Geospatial Analysis

Big data analytics has become a cornerstone in advancing remote sensing and geospatial analysis, enabling the efficient processing and interpretation of massive, complex datasets from multiple sources. By leveraging cloud computing, ML, and sophisticated data fusion techniques, researchers can extract valuable insights that were previously unattainable through traditional methods.

### 3.1. Processing and Storage Solutions

The unprecedented scale and complexity of remote sensing data in the big data era necessitate advanced processing and storage solutions to manage, analyze, and extract meaningful insights from vast and diverse geospatial datasets. Central to these advancements are cloud computing, edge computing, and distributed storage architectures, each of which plays a unique role in overcoming the technical challenges associated with data volume, velocity, and variety. Together, these solutions form the backbone of modern geospatial analytics, enabling researchers and practitioners to leverage the full potential of remote sensing data [40–42].

Cloud computing has revolutionized geospatial data processing by providing scalable, on-demand resources capable of handling massive datasets without the need for local infrastructure. Platforms like Amazon Web Services (AWS), Google Earth Engine, and Microsoft Azure allow users to store, access, and process petabytes of satellite, UAV, and IoT data, facilitating both real-time and batch processing workflows. The cloud's elasticity, combined with powerful data management and analytical tools, supports a broad range of geospatial applications, from complex ML models to real-time data fusion. Additionally, cloud platforms often provide integrated geospatial libraries and ML frameworks, streamlining processes that were previously labor-intensive, such as image classification, object detection, and change detection. This shift towards cloud-based geospatial analytics has democratized access to high-performance computing, enabling organizations of all sizes to perform advanced analyses and reduce infrastructure costs [43–45].

In parallel, edge computing has emerged as a critical solution for applications that require low-latency data processing close to the data source. Unlike cloud computing, which relies on centralized data centers, edge computing leverages localized computational resources at or near the data collection points, such as IoT sensors or UAVs. This distributed processing model is particularly valuable in dynamic, time-sensitive applications like disaster response, environmental monitoring, and autonomous navigation, where immediate insights are essential for decision making. By processing data on the edge, latency is minimized, bandwidth usage is optimized, and dependency on cloud connectivity is reduced, making edge computing a powerful tool for real-time geospatial analysis in remote or infrastructure-limited environments. The integration of edge devices with AI capabilities further enhances the potential of edge computing, enabling near-real-time image recognition, anomaly detection, and predictive analytics directly at the data source [46–48].

Complementing both cloud and edge computing is distributed storage, which addresses the challenges associated with managing large, complex, and often unstructured geospatial datasets. Traditional storage solutions are ill suited for big data geospatial applications, as they lack the scalability and flexibility needed to accommodate rapid data growth and heterogeneous data types. Distributed storage architectures, such as Hadoop Distributed File System (HDFS) and Apache Spark, offer robust data management frameworks that allow geospatial data to be stored and processed across multiple nodes, providing high fault tolerance and parallel processing capabilities. Data lakes, increasingly popular in geospatial analytics, are designed to store vast amounts of raw, unstructured data, including satellite imagery, sensor data, and environmental observations. Unlike traditional databases, data lakes support diverse data formats and schemas, making them highly adaptable to the heterogeneous nature of remote sensing data [49–52].

These processing and storage solutions, while powerful, also introduce challenges related to data transfer, security, and interoperability. The transmission of large datasets, especially from remote areas to centralized data centers, can be costly and time-consuming, underscoring the need for efficient data compression and bandwidth optimization techniques. Data security and privacy are also critical, particularly when dealing with sensitive geospatial information or proprietary sensor networks. Cloud providers have responded to these challenges with secure data encryption, access control, and compliance with global data privacy regulations, but these measures add layers of complexity that must be carefully managed to prevent data breaches and ensure user trust [53–56].

Furthermore, interoperability between cloud, edge, and distributed storage solutions is essential for a seamless geospatial analytics workflow. As data often originate from multiple sources and are processed across various platforms, standardizing data formats, metadata structures, and APIs is crucial to avoid data silos and enhance data accessibility. The adoption of open geospatial standards, such as those developed by the Open Geospatial Consortium (OGC), facilitates data exchange and integration, allowing researchers and practitioners to combine datasets from different platforms and derive more comprehensive insights [57–59].

The fusion of cloud computing, edge computing, and distributed storage forms a robust foundation for big data analytics in remote sensing, enabling the efficient management, processing, and analysis of large-scale geospatial datasets. These solutions, combined with advances in interoperability and data security, represent a paradigm shift in how geospatial information is processed and utilized. As remote sensing data grow in scale and diversity, these processing and storage solutions will be integral in unlocking new possibilities for real-time environmental monitoring, urban planning, disaster response, and other critical geospatial applications.

In conclusion, this subsection's exploration of advanced processing and storage solutions underscores the critical role of cloud computing, edge computing, and distributed storage in managing the vast scale and complexity of geospatial data in the big data era. Table 3 summarizes key references that delve into scalable architectures, low-latency processing, and robust data management frameworks to provide a concise overview of the foundational works and technological advancements discussed. This table serves as a guide for readers to further understand the state-of-the-art solutions driving modern geospatial analytics.

**Table 3.** Summary of references on advanced processing and storage solutions in big data remote sensing.

| Reference | Description | Specific Technology/Platform Used | Scalability Description |
|---|---|---|---|
| [40–42] | Overview of the role of cloud computing, edge computing, and distributed storage in addressing challenges of remote sensing data volume, velocity, and variety. | Google Cloud, Microsoft Azure | Highly scalable with distributed data centers and dynamic resource allocation. |
| [43–45] | Discuss scalable cloud computing platforms like Amazon Web Services, Google Earth Engine, and Azure for geospatial data processing and analysis, emphasizing elasticity, cost reduction, and integrated geospatial libraries. | AWS, Google Earth Engine | Elasticity allows for handling large geospatial workloads efficiently. |
| [46–48] | Highlight edge computing for low-latency, localized data processing near IoT sensors or UAVs, essential for time-sensitive geospatial applications like disaster response and environmental monitoring. | IoT Edge, UAV processing units | Limited scalability based on edge device constraints and processing power. |
| [49–52] | Explain distributed storage architectures and data lakes as scalable solutions for managing large, unstructured geospatial datasets. | HDFS, Apache Spark | Massively scalable for batch processing and fault-tolerant systems. |
| [53–56] | Discuss data transfer, security, and interoperability challenges in cloud, edge, and distributed storage, emphasizing the need for secure encryption, access control, and global privacy compliance. | Transport Layer Security (TLS) encryption, General Data Protection Regulation (GDPR) compliance tools | Scalability depends on encryption overhead and network infrastructure. |
| [57–59] | Explain the importance of open geospatial standards, such as those by OGC, for seamless data integration across diverse platforms and enhanced geospatial analytics. | OGC standards | Platform-independent scalability through open standards. |

### 3.2. Machine Learning and Artificial Intelligence

The integration of ML and AI into remote sensing and geospatial analysis has revolutionized the field. These methodologies enable sophisticated data processing, feature extraction, and predictive modeling capabilities that surpass traditional analytical methods. As geospatial datasets grow in size and complexity, ML provides the necessary tools to analyze, interpret, and extract actionable insights [60–63].

Recent advancements in CNNs have significantly enhanced the analysis of remote sensing imagery. CNNs excel at image classification, object detection, and segmentation tasks, which are critical for applications such as land cover mapping, urban infrastructure analysis, and natural resource monitoring. CNN-based models are widely preferred for their robustness in extracting spatial features from high-resolution satellite imagery and their ability to handle large-scale datasets. Compared to traditional methods, CNNs have demonstrated improved generalization and scalability, particularly in diverse environmental contexts [64–69].

Temporal data analysis has benefited from RNNs, particularly LSTM models. These models are adept at processing multi-temporal datasets, making them ideal for applications like monitoring vegetation cycles, predicting flood dynamics, and analyzing urban sprawl. LSTMs effectively integrate temporal dependencies and provide actionable forecasts for dynamic and time-sensitive applications. Their performance often surpasses that of traditional statistical methods due to their ability to handle non-linear temporal patterns in data [70–74].

Transfer learning has emerged as a practical strategy for addressing the challenges of data scarcity in EO. Pre-trained models, such as ResNet and Visual Geometry Group (VGG), are adapted to specialized geospatial datasets, enabling advanced analysis without the need for extensive labelled training data. Transfer learning approaches significantly reduce the computational and time costs of model training, facilitating broader adoption in resource-limited environments [75–79].

Explainable AI (XAI) techniques address one of the critical challenges of AI adoption in EO: model interpretability. Methods such as saliency maps and feature attribution provide transparency in the decision-making process of complex models. This transparency is particularly important in high-stakes applications, such as disaster management and urban planning, where trust in AI-driven decisions is paramount [80–83].

While specific numerical benchmarks may vary across datasets and applications, certain trends consistently emerge in the performance of AI methods for EO data processing. CNNs, for instance, have shown notable advantages in accuracy and feature extraction over traditional pixel-based methods, particularly in applications like land cover classification and object detection. LSTM models, with their ability to model temporal dependencies, consistently outperform linear statistical models in time-series predictions such as vegetation and flood dynamics [84–87].

Transfer learning demonstrates significant efficiency improvements, especially in scenarios where labelled data are scarce. Pre-trained models can be fine-tuned for geospatial tasks with minimal data, reducing training time and computational demands. Furthermore, XAI methodologies, while not directly enhancing predictive accuracy, improve the overall applicability of AI models by fostering trust and understanding among stakeholders [88–90].

Despite these advances, challenges related to computational efficiency, model scalability, and data heterogeneity persist. These challenges emphasize the need for hybrid AI approaches, where techniques like FL and dynamic model optimization can address both performance and resource constraints [91–93].

Overall, ML and AI represent powerful tools in the arsenal of remote sensing and geospatial analysis. By automating feature extraction, enhancing predictive capabilities, and supporting real-time decision making, AI and ML have transformed how geospatial data are understood and applied. As these technologies continue to evolve, their role in remote sensing will become even more integral, providing new avenues for research and applications in fields ranging from environmental monitoring to urban planning and disaster response [94–96].

In order to provide an organized view of the discussion, Table 4 provides a detailed classification of references, focusing on the integration of ML and AI in geospatial analysis. Each entry links a set of references and offers a comprehensive summary of their contributions to the field. Finally, Table 5 is based on real methodologies used in EO. This table summarizes the strengths, weaknesses, and optimal use cases for each method, drawing from the insights discussed in the subsection. The goal is to provide a concise reference tool for readers to understand the applicability of various AI techniques in EO tasks.

**Table 4.** ML and AI in geospatial analysis.

| References | Description | Scalability | Limitations |
|---|---|---|---|
| [60–63] | Overview of ML and AI integration in geospatial analysis, focusing on feature extraction and predictive modeling. | Scalability depends on the availability of cloud or distributed systems for handling large datasets. | Limited scalability in resource-constrained environments; high demand for computational resources. |
| [64–69] | Use of CNNs in remote sensing for image classification, object detection, and land cover mapping. | High scalability when paired with cloud computing platforms, though requires significant GPU/Tensor Processing Unit (TPU) support. | Computationally demanding; requires extensive labeled datasets for effective training. |
| [70–74] | Use of RNNs and LSTM for temporal data analysis, including vegetation monitoring and flood prediction. | Scalability limited by high computational costs and training time; resource-efficient alternatives needed for scaling. | Prone to overfitting and sensitive to hyperparameter tuning; computationally intensive for large datasets. |
| [75–79] | Application of transfer learning to address data scarcity in geospatial analysis. | Highly scalable in environments with limited labeled data due to reduced training requirements and model reuse. | Effectiveness depends on the relevance of pre-trained models to new domains; potential domain mismatches. |
| [80–83] | XAI methods for improving interpretability and transparency in AI models for geospatial applications. | Scalability improves trust and adoption in decision making but adds computational overhead for interpretability. | Does not directly improve predictive accuracy; computational overhead limits deployment in real-time applications. |
| [84–87] | Analysis of performance trends in AI models for EO, comparing accuracy and scalability. | Demonstrated scalability with cloud platforms and distributed frameworks for large-scale geospatial datasets. | High computational intensity for advanced models like CNNs; limited access to resources in low-infrastructure areas. |
| [88–90] | Efficiency improvements through transfer learning in scenarios with constrained data or resources. | Scalable due to reduced training times and computational demands, facilitating broader accessibility. | Challenges arise when adapting pre-trained models to highly specialized or divergent geospatial domains. |
| [91–96] | Identification of challenges in geospatial AI, such as computational efficiency, model scalability, and data heterogeneity, emphasizing hybrid approaches. | Scalability enhanced through hybrid techniques like FL and dynamic model optimization. | Data heterogeneity complicates model integration; resource requirements for training hybrid AI models are high. |

**Table 5.** Comparative summary of AI methods for EO data processing.

| AI Method | Strengths | Weaknesses | Optimal Use Cases |
|---|---|---|---|
| CNNs | - Excellent for spatial feature extraction and classification.<br>- Robust for handling high-resolution imagery.<br>- Can generalize well across diverse datasets. | - High computational demand, particularly with large datasets.<br>- Requires large labeled datasets for effective training. | - Land cover classification.<br>- Object detection in high-resolution satellite and UAV imagery.<br>- Urban infrastructure analysis. |
| RNNs–LSTM | - Ideal for temporal data analysis.<br>- Captures long-term dependencies in time-series data.<br>- Effective for sequential prediction tasks. | - Computationally intensive.<br>- Prone to overfitting if not carefully tuned.<br>- Requires significant training time. | - Vegetation monitoring.<br>- Flood prediction.<br>- Urban growth analysis. |

**Table 5.** *Cont.*

| AI Method | Strengths | Weaknesses | Optimal Use Cases |
|---|---|---|---|
| Transfer Learning | - Reduces computational time and data requirements.<br>- Enables model adaptation for specific tasks with limited data.<br>- Can leverage pre-trained models, improving performance in specialized domains. | - May not perform well if the pre-trained model is not sufficiently related to the new task.<br>- Potential domain mismatch when adapting models to new datasets. | - Tasks with limited labeled data.<br>- Quick adaptation to new geospatial datasets.<br>- Applications in resource-constrained environments. |
| XAI | - Increases model transparency and interpretability.<br>- Provides insights into model decision making, building trust in AI predictions.<br>- Crucial for high-impact applications where understanding model behavior is necessary. | - Does not directly improve predictive accuracy.<br>- May require additional computational overhead for interpretability techniques. | - Disaster response.<br>- Urban planning and decision making.<br>- Environmental monitoring where model transparency is critical. |

### 3.3. Data Fusion and Integration

Data fusion is a cornerstone of geospatial big data analysis, enabling the integration of heterogeneous datasets to generate comprehensive and actionable insights. As geospatial data sources continue to diversify, the ability to combine information from satellite imagery, UAVs, and IoT devices has become increasingly critical. Effective data fusion leverages the strengths of each data source while addressing their individual limitations, enhancing the reliability and utility of geospatial analysis [97–100].

Pixel-level fusion combines raw sensor data, such as optical satellite imagery and radar data, to enhance spatial, spectral, and temporal resolutions. For instance, high-resolution multispectral imagery fused with SAR data improves land use classification and vegetation mapping by capturing both surface reflectance and structural properties. Feature-level fusion extracts and combines key features from different datasets, enhancing interpretability and precision. UAV-acquired high-resolution imagery, for example, complements broader satellite-based observations, enabling detailed urban planning and infrastructure assessments. Decision-level fusion, which integrates outputs from multiple models or datasets, is particularly beneficial in scenarios like disaster response. For example, IoT-based flood sensors can validate satellite-derived flood maps, improving the accuracy of real-time response strategies [101–104].

The integration of IoT sensor networks with remote sensing data exemplifies hybrid approaches in geospatial analysis. IoT devices deliver localized, real-time measurements such as temperature, humidity, air quality, and soil moisture, which complement the extensive spatial coverage of satellite and UAV platforms. In precision agriculture, IoT sensors monitor micro-level field conditions while satellite-derived vegetation indices provide macroscopic insights. This combination enables optimized irrigation and fertilization strategies, reducing resource use while maximizing crop yields. In urban environments, hybrid approaches are increasingly applied to traffic management and pollution monitoring. IoT-enabled traffic sensors provide real-time data on congestion patterns, while UAV imagery captures dynamic changes in road conditions and urban activity. Integrating these data sources allows smart city initiatives to optimize traffic flows, reduce emissions, and enhance overall urban livability [105–108].

Dynamic data fusion techniques supported by AI and ML further enhance hybrid systems. These approaches adaptively weight the contributions of different data sources based on their reliability and relevance, enabling real-time adjustments in applications such as environmental monitoring and disaster management. AI-driven fusion models, for instance, can prioritize IoT sensor data during localized events such as floods while leveraging satellite imagery for broader situational awareness [109,110].

Despite these advancements, challenges persist in achieving seamless integration across heterogeneous datasets. Variations in spatial and temporal resolutions, differences in data formats, and the need for sensor calibration often complicate fusion processes. Cloud-based platforms and open-source tools, such as Google Earth Engine and Apache Spark, are increasingly employed to address these challenges by providing scalable and interoperable solutions. Data fusion and integration remain pivotal for unlocking the full potential of geospatial big data. By combining datasets from diverse sources and employing hybrid methodologies, these techniques enable more accurate and context-sensitive analysis, driving innovation in fields such as disaster management, precision agriculture, and smart city development [111–114].

In summary, data fusion and integration are transforming remote sensing and geospatial analysis by combining the strengths of diverse data sources to generate richer, more actionable insights. These techniques enable a holistic view of earth systems, allowing for precise monitoring and analysis responsive to spatial and temporal dynamics. As ML and cloud computing evolve, the potential for real-time, adaptive data fusion will expand, unlocking new applications and deeper insights in areas such as environmental monitoring, disaster response, and urban planning [115,116].

The key insights from this subsection are provided in Table 6, which presents an organized overview of the foundational references related to data fusion and integration in remote sensing. It focuses on data fusion and integration techniques in geospatial big data analysis. The table highlights the methodologies, challenges, and applications discussed in the references, including pixel-level, feature-level, and decision-level fusion, hybrid systems, and AI-driven integration approaches. Finally, Table 7 summarizes the key differences between traditional remote sensing methods and the transformative approaches enabled by big data analytics, illustrating their respective strengths, applications, and challenges.

**Table 6.** Data fusion and integration techniques in geospatial big data analysis.

| References | Description | Tools and Platforms Used | Limitations |
|---|---|---|---|
| [97–100] | Discusses the foundational role of data fusion in geospatial big data analysis, integrating diverse data sources such as satellite imagery, UAVs, and IoT devices. Pixel-level, feature-level, and decision-level fusion techniques are highlighted to address the limitations of individual data sources. | Google Earth Engine, Apache Spark | Variations in spatial and temporal resolutions; computational complexity in combining diverse data types. |
| [101–104] | Focuses on pixel-level and decision-level fusion techniques, with applications like SAR and optical image fusion. Discusses real-time disaster response scenarios and hybrid methods combining IoT data with remote sensing. | IoT Edge frameworks, Hybrid cloud systems | Real-time data synchronization challenges; high latency in remote areas without edge infrastructure. |

**Table 6.** *Cont.*

| References | Description | Tools and Platforms Used | Limitations |
|---|---|---|---|
| [105–108] | Describes hybrid mist–cloud systems for large-scale geospatial data analytics, smart city applications, and traffic management using IoT and UAV data. | Cloud-based platforms (AWS, Microsoft Azure), UAV frameworks | Scalability issues for resource-intensive applications; dependency on high-bandwidth infrastructure. |
| [109,110] | Highlights dynamic data fusion techniques using AI/ML for adaptive weighting of heterogeneous data sources, enabling real-time environmental and disaster monitoring. | Dynamic AI frameworks, ML Pipelines | High computational demands; limited interpretability of AI-driven fusion models. |
| [111–114] | Explains challenges in achieving seamless integration of heterogeneous datasets and emphasizes the role of cloud-based platforms and open-source tools for scalable solutions. | Open-source tools (e.g., Apache Spark, Hadoop), distributed storage systems | Data heterogeneity and format inconsistencies; lack of standardization across platforms. |
| [115,116] | Summarizes the transformative potential of real-time and adaptive data fusion methodologies for environmental monitoring, disaster response, and urban planning. | Real-time IoT networks, edge computing | Latency in integrating multi-source data streams; dependency on robust sensor networks. |

**Table 7.** Comparison of traditional vs. big data-enabled geospatial methods.

| Aspect | Traditional Methods | Big Data-Enabled Methods |
|---|---|---|
| Data Volume | Limited datasets collected periodically from single sources like satellites or aerial imagery. | Massive, heterogeneous datasets from satellites, UAVs, IoT sensors, and social media, reaching petabyte scales. |
| Data Processing | Manual, labor-intensive workflows using desktop geographic information system (GIS) software . | Automated workflows leveraging cloud platforms (e.g., Google Earth Engine) and AI/ML pipelines. |
| Temporal Resolution | Fixed intervals based on satellite revisit times (e.g., daily, weekly). | Near real-time data streams from IoT networks and satellite constellations (e.g., Planet Labs, Sentinel-2). |
| Data Integration | Limited, often manual integration of multi-source datasets. | Advanced data fusion techniques (e.g., pixel-, feature-, and decision-level integration using ML algorithms). |
| Analytical Techniques | Rule-based models, statistical techniques, and basic image classification. | AI-driven approaches (e.g., CNNs for image analysis, RNNs for temporal data, and transfer learning for specialized tasks). |
| Applications | Mapping, static land cover classification, and historical trend analysis. | Predictive analytics, real-time monitoring, disaster response, urban planning, precision agriculture, and climate change modeling. |
| Scalability | Limited scalability due to reliance on local hardware and software. | Scalable solutions leveraging cloud computing (AWS, Google Cloud) and distributed storage systems (HDFS, Apache Spark). |
| Accessibility | Restricted access due to high costs of proprietary software and limited computational resources. | Democratized access via open-source platforms, cloud services, and data-sharing initiatives (e.g., Copernicus Open Access Hub). |
| Challenges | Constrained by data volume, storage capacity, and manual workflows. | Ethical and technical challenges including data privacy, interoperability, and bias in AI models. |

## 4. Applications in Remote Sensing and Geospatial Analysis

Remote sensing and geospatial analysis have become indispensable tools across a wide range of fields, as they enable detailed monitoring, forecasting, and management of environmental, urban, agricultural, and health systems. By integrating diverse data sources with advanced analytical methods, these applications address complex challenges in ways that were previously unattainable. One of the most vital applications is environmental monitoring, where high-resolution satellite and UAV imagery provide insights into terrestrial and aquatic ecosystems. Continuous observation of forests, wetlands, and marine environments enables scientists to track vegetation health, wildlife habitats, and land use changes over time. Data fusion techniques that combine optical, Light Detection and Ranging (LiDAR), and SAR data add a multidimensional perspective to environmental analysis, distinguishing between nuanced changes such as species migration, forest degradation, or coastal erosion. These insights are crucial for conservation efforts, climate change adaptation strategies, and understanding ecological dynamics affected by human activities [117–120].

In the realm of disaster management, remote sensing offers critical support by providing rapid and accurate data essential for planning, response, and recovery. Following natural disasters like hurricanes, earthquakes, floods, or wildfires, high-resolution imagery from satellites and UAVs aids in assessing damage extent, identifying affected regions, and planning resource deployment. Integrating multi-temporal data enables change detection, which is vital for prioritizing rescue and relief efforts in the immediate aftermath. ML models applied to these images can detect features like collapsed buildings, inundated areas, or active fire perimeters, enhancing real-time awareness and facilitating swift responses. Moreover, historical data and predictive models allow for proactive disaster preparedness, guiding infrastructure planning in high-risk zones and supporting early warning systems that minimize potential risks [121–124].

Urban planning and the development of smart cities are also significantly enhanced by remote sensing, which provides critical insights into land use, infrastructure growth, and population density. High-resolution imagery and spatial analysis help urban planners monitor city expansion, assess land cover changes, and evaluate urbanization impacts on the environment. These insights are central to smart city initiatives, which rely on data-driven approaches for optimizing energy use, transportation systems, waste management, and other urban resources. Remote sensing further supports the assessment of urban green spaces, pollution levels, and heat islands, contributing to healthier, more resilient urban environments. The integration of IoT data with remote sensing in smart cities allows for real-time monitoring, supporting responsive management of urban challenges like traffic congestion and air quality [125–130].

In agriculture, remote sensing has driven the evolution of precision farming, allowing farmers to optimize crop production, reduce waste, and make more sustainable choices. Multispectral and hyperspectral imagery captures fine details about crop health, soil moisture, and nutrient levels, helping farmers make data-informed decisions on irrigation, fertilization, and pest control. UAVs equipped with sensors provide high-resolution, field-specific information, highlighting areas requiring intervention and enabling real-time monitoring of crop conditions. ML models further enhance agricultural practices by analyzing historical data to identify patterns in crop health and predict future yields based on environmental factors. The use of remote sensing in agriculture reduces resource use, lowers costs, and mitigates the environmental impact of farming practices [131–135].

Remote sensing also plays an indispensable role in climate change research by providing data essential for studying global climate patterns and the impacts of human activities on Earth's systems. Satellite sensors capture data on sea surface temperature, ice sheet volume, and greenhouse gas concentrations, offering long-term perspectives on climate dynamics. By integrating multi-temporal and multi-spectral data, researchers can track changes in polar ice, sea levels, and vegetation, which are critical indicators of climate health. These data enable scientists to assess the relationship between climate change and extreme weather events like hurricanes and droughts, enhancing our understanding of how shifting climate patterns impact global ecosystems. ML models are increasingly applied to predict climate outcomes, helping policymakers anticipate and mitigate the adverse effects of climate change [136–140].

The management of water resources also benefits significantly from remote sensing, as it allows continuous monitoring of freshwater sources, watershed areas, and groundwater levels. Remote sensing technologies can track water quality indicators, such as turbidity and chlorophyll concentrations, which are essential for water resource planning and pollution management. SAR and optical data are used to monitor surface water changes, aiding in flood forecasting and drought assessment. This information is particularly valuable in water-scarce regions and densely populated urban areas, where managing water resources efficiently is critical for public health and economic stability [141–144].

In public health and epidemiology, remote sensing is emerging as a valuable tool for tracking and predicting the spread of infectious diseases. Mapping vegetation, water bodies, and population density helps identify conditions conducive to disease vectors, such as mosquitoes carrying malaria and dengue. ML models applied to remote sensing data predict disease outbreak likelihood based on environmental and climatic factors, allowing for targeted public health interventions. Geospatial analysis in urban areas also assesses factors such as air quality, green space, and population density, directly impacting public health outcomes [145–149].

In summary, remote sensing and geospatial analysis serve as powerful, data-driven tools that enhance our ability to monitor, manage, and respond to diverse environmental, urban, agricultural, and health challenges. The integration of big data analytics, ML, and multi-source data fusion allows for a holistic and real-time understanding of dynamic systems, driving informed decision making and fostering resilience across disciplines. As these technologies evolve, their applications will continue to expand, offering new opportunities for scientific discovery and actionable insights.

Table 8 compiles the essential references discussed in this section, categorizing the driving advancements in fields such as environmental monitoring, disaster management, urban planning, agriculture, climate research, water resource management, and public health. This comprehensive resource highlights the breadth and impact of remote sensing technologies across various domains.

**Table 8.** Key references supporting diverse applications of remote sensing and geospatial analysis.

| References | Description | Tools and Techniques | Limitations |
|---|---|---|---|
| [117–120] | Focus on environmental monitoring, tracking vegetation health, and land use changes over time. | Optical sensors, LiDAR, SAR, geospatial analytics platforms like Google Earth Engine | Data heterogeneity; processing challenges for large-scale temporal datasets. |
| [121–124] | Disaster management applications, including damage assessment and predictive risk modeling. | Multi-temporal data fusion, UAV platforms, AI/ML for predictive analytics | Real-time data processing challenges; dependency on high-resolution imagery for accuracy. |
| [125–130] | Urban planning and smart city development using high-resolution imagery for infrastructure analysis. | IoT-based monitoring systems, cloud platforms (AWS, Azure), GIS tools | Scalability issues with IoT systems; high latency in large-scale urban environments. |

**Table 8.** *Cont.*

| References | Description | Tools and Techniques | Limitations |
|---|---|---|---|
| [131–135] | Applications in precision agriculture, utilizing multispectral and hyperspectral imagery. | UAV-based field monitoring, crop analysis algorithms | Limited field coverage in real time; reliance on weather conditions for UAV operations. |
| [136–140] | Contributions to climate change research, including sea surface temperature and ice sheet monitoring. | Satellite platforms (MODIS, Landsat), DL models for temporal and spectral analysis | Computational demands for global-scale modeling; limited resolution of some datasets. |
| [141–144] | Water resource management, focusing on monitoring water quality and watershed areas. | SAR and optical imaging systems, geospatial hydrology models | Calibration challenges for multi-sensor systems; resource-intensive modeling of water dynamics. |
| [145–149] | Public health and epidemiology applications, such as tracking infectious diseases and air quality. | Air quality sensors, spatial epidemiology frameworks, real-time data streams | Data security issues for health datasets; integration complexity of spatial and epidemiological data. |

## 5. Case Studies and Application Domains

Flood risk assessment has significantly advanced through the integration of geospatial big data and analytical frameworks. Satellite imagery, particularly from programs like Sentinel-1 and Sentinel-2, is frequently used to map flood-prone regions. ML models enhance flood extent prediction by analyzing multi-temporal data and hydrological factors. Combined with IoT sensors for real-time water level monitoring, these methods provide a comprehensive approach to flood mitigation, enabling dynamic risk modeling and resource allocation.

Urban expansion requires precise monitoring to manage land use and optimize infrastructure. SAR data are used to detect changes in urban surfaces, while UAV imagery supports the identification of construction activity and population density changes. These datasets, when integrated using data fusion techniques, allow urban planners to model transportation networks, evaluate housing needs, and manage green spaces. Big data analytics enables the analysis of trends over time, supporting sustainable development policies.

Precision agriculture benefits from UAV-mounted multispectral sensors and satellite imagery to monitor crop health, soil moisture, and nutrient distribution. Advanced data fusion integrates historical climate data and real-time weather forecasts to optimize farming practices. These methods reduce water and fertilizer usage while maximizing crop yields, demonstrating the transformative potential of big data in agricultural management.

The detection of environmental changes, such as deforestation, desertification, or glacier retreat, is an essential application of geospatial big data. Hyperspectral imagery and SAR data allow detailed mapping of vegetation and terrain changes over time. Combined with DL algorithms, these datasets support early warnings for environmental degradation and inform conservation policies.

Big data analytics supports the optimization of transportation and logistics networks by combining traffic monitoring, geospatial mapping, and predictive analytics. AI-based models trained on historic traffic patterns and real-time Global Positioning System (GPS) data help minimize congestion and reduce fuel consumption. These insights are crucial for urban logistics, fleet management, and smart city initiatives.

# 6. Challenges and Future Directions

As remote sensing and geospatial analysis advance, the field faces significant challenges related to data management, interoperability, and model interpretability. However, emerging technologies and innovative methodologies hold immense promise to address these obstacles, paving the way for transformative applications and more sophisticated analytics.

## *6.1. Challenges*

As remote sensing and geospatial analysis evolve within the big data landscape, managing the exponential growth of geospatial data continues to be a significant challenge. The sheer volume of data generated by satellites, UAVs, and IoT devices necessitates robust and scalable architectures for storage, processing, and analysis. These challenges are compounded by the need for real-time insights, especially in dynamic applications like disaster response and urban monitoring, where computational efficiency becomes critical [150–154].

High-resolution, multi-source geospatial data present substantial computational demands, particularly when using AI methodologies such as DL models. CNNs for image analysis or LSTM models for temporal predictions require significant computational power, often necessitating specialized hardware such as GPUs or Tensor Processing Units (TPUs). This high demand for resources creates disparities in access to advanced analytics, particularly for resource-constrained organizations or regions. The computational inefficiencies in handling hyperspectral imagery and multi-temporal data further exacerbate this issue, requiring optimization of algorithms and data pipelines [155–158].

The reliance on cloud computing and high-performance computing systems has alleviated some of these challenges. However, the associated costs, bandwidth limitations, and latency concerns remain barriers to widespread adoption. To address these issues, edge computing has emerged as a promising solution for decentralized data processing. By performing computations near the data source, edge computing reduces latency and alleviates bandwidth constraints, making it ideal for time-sensitive applications like flood monitoring or urban traffic optimization [159–162].

AI methodologies have begun addressing computational challenges through techniques such as model compression, pruning, and quantization. These approaches reduce model complexity without compromising accuracy, enabling faster inference and lower resource consumption. FL also offers a potential pathway for improving computational efficiency by distributing the training workload across decentralized nodes, reducing the dependency on centralized computational infrastructure [163–165].

Despite these advancements, challenges related to data heterogeneity, interoperability, and ethical concerns persist. Variations in spatial and temporal resolutions across datasets complicate integration and analysis, while the lack of standardized formats increases the effort required for pre-processing. Ethical issues, particularly in data privacy and AI transparency, further complicate the implementation of AI-driven methodologies in EO [166–168].

Table 9 focuses on challenges in geospatial big data analysis. It categorizes references by their key contributions to topics such as data management, computational efficiency, scalability, real-time processing, and ethical considerations. The table emphasizes the advancements in AI methodologies, edge computing, and FL while addressing critical issues like data heterogeneity and privacy. It serves as a concise resource for understanding the complexities and opportunities in managing geospatial data at scale.

**Table 9.** Challenges and opportunities in geospatial big data.

| References | Description | Solutions Adopted | Limitations and Considerations |
|---|---|---|---|
| [150–154] | Challenges in managing the exponential growth of geospatial data from diverse sources. | Use of cloud computing for scalable storage and processing; distributed frameworks for handling large data. | High bandwidth requirements for real-time data; computational inefficiencies in large-scale systems. |
| [155–158] | Computational demands of high-resolution, multi-source data, especially for AI methods like CNNs. | Model compression, pruning, and edge computing to optimize resource usage. | Specialized hardware like GPUs/TPUs required; high costs and limited accessibility for resource-constrained users. |
| [159–162] | Bandwidth and latency issues in decentralized data processing for real-time applications. | Edge computing near data sources to reduce latency and optimize bandwidth. | Dependency on localized infrastructure; limited scalability in remote regions. |
| [163–165] | Computational inefficiencies in AI-driven geospatial models and scalability issues. | FL for decentralized training and dynamic model optimization. | Data heterogeneity across nodes; potential model bias due to unequal data distributions. |
| [166–168] | Challenges related to data heterogeneity, interoperability, and ethical concerns in geospatial AI. | Adoption of open standards for data formats and secure aggregation protocols in FL. | Lack of global standardization; privacy concerns in sensitive datasets and AI decision-making transparency. |

### 6.2. Future Directions

Despite these challenges, the future of remote sensing and geospatial analysis is promising, as emerging technologies and methodological advancements are poised to reshape the field. Quantum computing represents one of the most exciting frontiers, with the potential to dramatically accelerate data processing speeds and tackle complex optimization problems inherent in geospatial analysis. By leveraging the principles of quantum mechanics, quantum computers could enable rapid processing of vast datasets, particularly beneficial for high-dimensional geospatial data, such as hyperspectral imagery. For instance, Grover's algorithm could significantly enhance spatial search efficiency, enabling faster identification of specific land cover types or anomalies in hyperspectral datasets. Similarly, quantum annealing techniques can optimize the scheduling of satellite imaging tasks, ensuring better resource utilization in real-time applications such as disaster response. Quantum computing could also address combinatorial optimization challenges, such as designing efficient routing for UAVs in dynamic environments or optimizing resource allocation in multi-satellite constellations [169–172].

Several research initiatives, such as IBM's Qiskit and Google's Sycamore project, are exploring the application of quantum algorithms to geospatial challenges, including environmental monitoring and predictive modeling. Emerging collaborations between quantum computing and remote sensing research communities highlight the potential for significant breakthroughs in processing massive EO datasets. However, the current scalability of quantum hardware and the need for error correction remain significant barriers to its widespread adoption. Hybrid quantum–classical systems may provide a transitional solution for near-term geospatial applications [173–176].

FL is another innovative approach that could revolutionize remote sensing by enabling collaborative model training without requiring centralized data storage. FL offers a powerful solution to address data privacy and security concerns in geospatial analysis by enabling collaborative model training across multiple decentralized datasets without transferring sensitive data to a central repository. For instance, disaster management agencies from different regions can collaboratively train predictive models for flood or wildfire spread without exposing proprietary or sensitive local datasets. Similarly, in urban planning, FL can integrate IoT sensor data from various cities to develop traffic optimization models or monitor pollution levels while adhering to stringent privacy regulations. By decentralizing the learning process, FL not only safeguards data privacy but also minimizes the risks of data breaches, making it particularly valuable for applications involving confidential or sensitive geospatial information [177–180].

Several frameworks, such as TensorFlow Federated and PySyft, have been developed to facilitate the implementation of FL in geospatial domains. These tools allow the seamless integration of decentralized datasets from satellites, UAVs, and IoT devices into a unified training process while maintaining data localization. FL also reduces the computational burden on central servers by distributing the training load to edge devices, making it ideal for resource-constrained environments. For example, an FL framework could enable real-time updates to environmental monitoring models using IoT data collected from rural or remote areas. Furthermore, ongoing research into secure aggregation protocols ensures that the outcomes of FL remain accurate and unbiased, even when trained across heterogeneous datasets [181–184].

Enhanced data fusion techniques represent another key area of future development, as they enable more sophisticated integration of multi-source data. Advances in DL and AI are allowing for dynamic, adaptive fusion methods that can combine data from different sources and resolutions based on the specific requirements of a given application. For instance, next-generation data fusion techniques might allow real-time integration of satellite, UAV, and ground sensor data, providing a comprehensive, continuously updated view of complex environments. This adaptability would be particularly useful for applications such as disaster management, where rapidly changing conditions require a high degree of responsiveness. Continued research into AI-driven data fusion is expected to further improve the accuracy and timeliness of multi-source analyses [185–188].

The development of real-time geospatial analytics is also a growing priority, particularly in applications where timely data are critical for decision making. The rise of edge computing, which involves processing data near the data source rather than in a centralized data center, holds promise for enabling real-time geospatial analytics in remote and resource-constrained settings. Edge computing reduces latency and conserves bandwidth by minimizing the need to transmit large datasets across networks, making it an ideal solution for applications like autonomous navigation, disaster response, and environmental monitoring. As IoT infrastructure expands and edge computing devices become more powerful, the field is likely to see significant advances in real-time geospatial capabilities [189–192].

Furthermore, the integration of XAI into geospatial analytics is expected to improve the transparency and interpretability of AI models, addressing concerns about black-box algorithms in critical applications. Future research in XAI may lead to the development of models that not only produce accurate predictions but also provide interpretable, user-friendly explanations of their decision-making processes. This transparency will be crucial in building trust with stakeholders and ensuring that AI-based insights are used responsibly in geospatial analysis [193–195].

Lastly, interdisciplinary collaboration and open data initiatives will play a crucial role in advancing remote sensing and geospatial analysis. As more data become available from government agencies, academic institutions, and private entities, fostering open data sharing and collaboration across sectors will be essential for addressing global challenges such as climate change, disaster preparedness, and urban sustainability. The establishment of open-access geospatial data repositories and standardized data-sharing protocols will facilitate broader access to geospatial data, democratizing research and enabling a more comprehensive approach to global monitoring and analysis [196–198].

In summary, the future of remote sensing and geospatial analysis is marked by significant technological advancements and the potential for transformative impact across multiple domains. While challenges in data management, privacy, and model interpretability persist, emerging technologies such as quantum computing, FL, and XAI hold promise for overcoming these obstacles. By embracing these innovations and fostering collaboration, the field is poised to unlock new insights and solutions that will reshape how we understand and manage our planet's dynamic systems. Table 10 focuses on future directions in geospatial analysis. It categorizes key advancements, including quantum computing for optimization challenges, FL for privacy-preserving decentralized training, adaptive data fusion techniques for real-time multi-source integration, and edge computing for real-time geospatial analytics. Additionally, it addresses the role of XAI in enhancing transparency and trust in AI-driven geospatial models. The table also highlights the importance of interdisciplinary collaboration and open data initiatives in advancing global geospatial research and tackling critical challenges like climate change and disaster preparedness.

**Table 10.** Future directions in geospatial analysis.

| Reference | Description | Applications and Impact | Challenges in Implementation |
|---|---|---|---|
| [169–172] | Explores quantum computing applications in geospatial analysis, focusing on optimization problems like satellite imaging schedules, UAV routing, and resource allocation. Highlights algorithms like Grover's for efficient spatial search and quantum annealing for combinatorial optimization. | Optimization of satellite task schedules, efficient UAV routing for emergency response, resource allocation in constrained geospatial scenarios. | High computational cost, complexity in quantum algorithm design, limited availability of quantum hardware. |
| [173–176] | Details research into hybrid quantum–classical systems and error correction in quantum computing, emphasizing their potential to accelerate geospatial data processing for tasks like disaster response and climate modeling. | Faster geospatial data analysis for disaster response, improved accuracy in climate modeling predictions. | Error rates in quantum systems, high energy consumption, integration challenges with existing systems. |
| [177–180] | Discusses FL for privacy-preserving, decentralized training in geospatial applications, such as urban planning, disaster management, and collaborative environmental monitoring. | Collaborative urban planning, enhanced disaster response strategies, real-time monitoring of environmental factors while preserving data privacy. | Data heterogeneity across nodes, unequal distribution of resources, communication overhead in federated systems. |
| [181–184] | Highlights edge computing solutions for real-time geospatial analytics in remote areas, reducing latency and bandwidth usage. Discusses adaptive data fusion techniques for multi-source integration. | Real-time geospatial data processing in remote locations, adaptive response to environmental changes, enhanced decision-making speed. | Infrastructure constraints in remote regions, energy consumption in edge devices, scalability issues for large-scale deployment. |
| [185–188] | Focuses on the integration of XAI in geospatial applications to improve transparency and trust in AI models. | Transparent AI-driven decision making, improved stakeholder trust in urban planning and disaster management. | Difficulty in balancing model complexity with interpretability, lack of standardized XAI frameworks for geospatial analysis. |

**Table 10.** *Cont.*

| Reference | Description | Applications and Impact | Challenges in Implementation |
| --- | --- | --- | --- |
| [189–192] | Stresses the role of interdisciplinary collaboration and open data initiatives in advancing global geospatial research, tackling challenges like climate change and disaster preparedness. | Democratization of geospatial research, enhanced global collaboration on climate change mitigation strategies. | Variability in data quality, privacy concerns in open-access data sharing, and harmonization of diverse data formats. |
| [193–195] | Emphasizes the importance of XAI for building trust in AI systems by enhancing transparency in critical geospatial applications. | Ensures accountability in geospatial decision making, such as disaster recovery strategies and resource allocation. | Ethical concerns about AI-driven decisions and insufficient adoption of XAI principles in global systems. |
| [196–198] | Highlights interdisciplinary collaboration and open data initiatives to address global challenges such as climate change and disaster preparedness. | Democratized access to geospatial data repositories, fostering global collaboration on environmental sustainability. | Issues with standardizing data-sharing protocols and ensuring the security of shared datasets. |

## 7. Conclusions

This survey has explored the pivotal role of big data analytics in revolutionizing remote sensing and geospatial analysis. As the volume and complexity of geospatial data grow, traditional methods are increasingly complemented by advanced AI-driven techniques such as ML, DL, and data fusion, enabling more precise and real-time insights into Earth's dynamic systems. The integration of multi-source data from satellites, UAVs, and IoT sensors has significantly enhanced the scope of geospatial analysis, allowing for improved monitoring and decision making in domains ranging from environmental monitoring to urban planning.

AI methodologies such as CNNs and LSTM networks have shown considerable promise in applications involving spatial and temporal data, respectively. These methods offer enhanced predictive accuracy, particularly when processing large-scale and multi-temporal datasets. Furthermore, transfer learning has emerged as a critical strategy for overcoming challenges associated with limited labeled data, making advanced AI techniques more accessible in resource-constrained environments.

Despite these advancements, the integration of AI in geospatial analysis faces significant challenges. Issues related to computational efficiency, data interoperability, and model interpretability remain at the forefront. However, the adoption of cloud-based computing, edge processing, and XAI techniques is gradually addressing these obstacles, ensuring that models not only deliver high performance but also maintain transparency and trustworthiness.

Looking forward, the continued development of hybrid approaches that combine AI with traditional methods, alongside further innovations in cloud computing and distributed data processing, will drive the next phase of geospatial analysis. As these technologies evolve, their ability to tackle global challenges—such as climate change, urbanization, and natural disasters—will grow, solidifying the role of remote sensing as an indispensable tool in the pursuit of sustainable development and informed decision making.

In conclusion, the integration of big data analytics into remote sensing and geospatial analysis is poised to significantly enhance our understanding of complex Earth systems. The combination of cutting-edge AI methodologies with diverse data sources offers unprecedented opportunities to monitor, manage, and predict environmental and urban dynamics, paving the way for smarter, more sustainable futures.

**Author Contributions:** E.D. and M.T. conceived of the idea, designed and performed the experiments, analyzed the results, drafted the initial manuscript and revised the final manuscript. All authors have read and agreed to the published version of the manuscript.

**Funding:** This research received no external funding.

**Conflicts of Interest:** The authors declare no conflicts of interest.

## References

1. Dian, R.; Li, S.; Sun, B.; Guo, A. Recent advances and new guidelines on hyperspectral and multispectral image fusion. *Inf. Fusion* **2021**, *69*, 40–51. [CrossRef]
2. Ustin, S.L.; Middleton, E.M. Current and near-term advances in Earth observation for ecological applications. *Ecol. Process.* **2021**, *10*, 1. [CrossRef] [PubMed]
3. Breunig, M.; Bradley, P.E.; Jahn, M.; Kuper, P.; Mazroob, N.; Rösch, N.; Al-Doori, M.; Stefanakis, E.; Jadidi, M. Geospatial data management research: Progress and future directions. *ISPRS Int. J. Geo-Inf.* **2020**, *9*, 95. [CrossRef]
4. Rai, P.K.; Mishra, V.N.; Singh, P. *Geospatial Technology for Landscape and Environmental Management*; Springer: Berlin/Heidelberg, Germany, 2022.
5. Roscher, R.; Rußwurm, M.; Gevaert, C.; Kampffmeyer, M.; dos Santos, J.A.; Vakalopoulou, M.; Hänsch, R.; Hansen, S.; Nogueira, K.; Prexl, J.; et al. Data-centric machine learning for geospatial remote sensing data. *arXiv* **2023**, arXiv:2312.05327.
6. Liang, M.; Li, H.; Liu, R.W.; Lam, J.S.L.; Yang, Z. PiracyAnalyzer: Spatial temporal patterns analysis of global piracy incidents. *Reliab. Eng. Syst. Saf.* **2024**, *243*, 109877. [CrossRef]
7. Chen, J.; Chen, S.; Fu, R.; Li, D.; Jiang, H.; Wang, C.; Peng, Y.; Jia, K.; Hicks, B.J. Remote sensing big data for water environment monitoring: Current status, challenges, and future prospects. *Earth's Future* **2022**, *10*, e2021EF002289. [CrossRef]
8. Wang, L.; Yan, J.; Ma, Y. *Cloud Computing in Remote Sensing*; Chapman and Hall/CRC: Boca Raton, FL, USA, 2019.
9. Islam, H.; Abbasi, H.; Karam, A.; Chughtai, A.H.; Ahmed Jiskani, M. Geospatial analysis of wetlands based on land use/land cover dynamics using remote sensing and GIS in Sindh, Pakistan. *Sci. Prog.* **2021**, *104*, 00368504211026143. [CrossRef]
10. Dong, J.; Metternicht, G.; Hostert, P.; Fensholt, R.; Chowdhury, R.R. Remote sensing and geospatial technologies in support of a normative land system science: Status and prospects. *Curr. Opin. Environ. Sustain.* **2019**, *38*, 44–52. [CrossRef]
11. Haq, B.U.; Faisal, M.; Haq, M.U. Ethical Considerations in Remote Sensing and Cognitive Modeling. In *Deep Cognitive Modelling in Remote Sensing Image Processing*; IGI Global: Hershey, PA, USA, 2024; pp. 70–87.
12. Hong, D.; He, W.; Yokoya, N.; Yao, J.; Gao, L.; Zhang, L.; Chanussot, J.; Zhu, X. Interpretable hyperspectral artificial intelligence: When nonconvex modeling meets hyperspectral remote sensing. *IEEE Geosci. Remote Sens. Mag.* **2021**, *9*, 52–87. [CrossRef]
13. Wang, S.; Han, W.; Zhang, X.; Li, J.; Wang, L. Geospatial remote sensing interpretation: From perception to cognition. *Innov. Geosci.* **2024**, *2*, 100056. [CrossRef]
14. Chuvieco, E. *Fundamentals of Satellite Remote Sensing: An Environmental Approach*; CRC Press: Boca Raton, FL, USA, 2020.
15. Elachi, C.; Van Zyl, J.J. *Introduction to the Physics and Techniques of Remote Sensing*; John Wiley & Sons: Hoboken, NJ, USA, 2021.
16. Feroz, S.; Abu Dabous, S. Uav-based remote sensing applications for bridge condition assessment. *Remote Sens.* **2021**, *13*, 1809. [CrossRef]
17. Wang, B.; Lu, X.; Zheng, X.; Li, X. Semantic descriptions of high-resolution remote sensing images. *IEEE Geosci. Remote Sens. Lett.* **2019**, *16*, 1274–1278. [CrossRef]
18. Polonelli, T.; Qin, Y.; Yeatman, E.M.; Benini, L.; Boyle, D. A flexible, low-power platform for UAV-based data collection from remote sensors. *IEEE Access* **2020**, *8*, 164775–164785. [CrossRef]
19. Hooda, Y. IoT and remote sensing. In *Computer Vision and Internet of Things*; Chapman and Hall/CRC: Boca Raton, FL, USA, 2022; pp. 111–140.
20. Venter, Z.S.; Brousse, O.; Esau, I.; Meier, F. Hyperlocal mapping of urban air temperature using remote sensing and crowdsourced weather data. *Remote Sens. Environ.* **2020**, *242*, 111791. [CrossRef]
21. Cheng, Y.; Zhou, K.; Wang, J.; Yan, J. Big earth observation data integration in remote sensing based on a distributed spatial framework. *Remote Sens.* **2020**, *12*, 972. [CrossRef]
22. Wu, X.; Li, W.; Hong, D.; Tian, J.; Tao, R.; Du, Q. Vehicle detection of multi-source remote sensing data using active fine-tuning network. *ISPRS J. Photogramm. Remote Sens.* **2020**, *167*, 39–53. [CrossRef]
23. Zipkin, E.F.; Zylstra, E.R.; Wright, A.D.; Saunders, S.P.; Finley, A.O.; Dietze, M.C.; Itter, M.S.; Tingley, M.W. Addressing data integration challenges to link ecological processes across scales. *Front. Ecol. Environ.* **2021**, *19*, 30–38. [CrossRef]

24. Sun, W.; Zhou, J.; Meng, X.; Yang, G.; Ren, K.; Peng, J. Coupled temporal variation information estimation and resolution enhancement for remote sensing spatial–temporal–spectral fusion. *IEEE Trans. Geosci. Remote Sens.* **2023**, *61*, 5532118. [CrossRef]
25. Qi, X.; Zhu, P.; Wang, Y.; Zhang, L.; Peng, J.; Wu, M.; Chen, J.; Zhao, X.; Zang, N.; Mathiopoulos, P.T. MLRSNet: A multi-label high spatial resolution remote sensing dataset for semantic scene understanding. *ISPRS J. Photogramm. Remote Sens.* **2020**, *169*, 337–350. [CrossRef]
26. Lawhead, J. *Learning Geospatial Analysis with Python: Understand GIS Fundamentals and Perform Remote Sensing Data Analysis Using Python 3.7*; Packt Publishing Ltd.: Birmingham, UK, 2019.
27. Sabins, F.F., Jr.; Ellis, J.M. *Remote Sensing: Principles, Interpretation, and Applications*; Waveland Press: Long Grove, IL, USA, 2020.
28. Cui, Y.; Chen, X.; Xiong, W.; He, L.; Lv, F.; Fan, W.; Luo, Z.; Hong, Y. A soil moisture spatial and temporal resolution improving algorithm based on multi-source remote sensing data and GRNN model. *Remote Sens.* **2020**, *12*, 455. [CrossRef]
29. Materazzi, F.; Pacifici, M.; Santaga, F.S. From top to bottom. Multispectral Remote Sensing and Data Integration to Rediscover Veii. *FOLD&R* **2024**. Available online: https://www.fastionline.org/docs/FOLDER-it-2024-577.pdf (accessed on 15 December 2024)
30. Wang, W.; Shi, K.; Zhang, Y.; Li, N.; Sun, X.; Zhang, D.; Zhang, Y.; Qin, B.; Zhu, G. A ground-based remote sensing system for high-frequency and real-time monitoring of phytoplankton blooms. *J. Hazard. Mater.* **2022**, *439*, 129623. [CrossRef] [PubMed]
31. Stamford, J.; Aciksoz, S.B.; Lawson, T. Remote sensing techniques: Hyperspectral imaging and data analysis. In *Photosynthesis: Methods and Protocols*; Springer: Berlin/Heidelberg, Germany, 2024; pp. 373–390.
32. Zhao, J.; Zhong, Y.; Hu, X.; Wei, L.; Zhang, L. A robust spectral-spatial approach to identifying heterogeneous crops using remote sensing imagery with high spectral and spatial resolutions. *Remote Sens. Environ.* **2020**, *239*, 111605. [CrossRef]
33. Richards, J.A.; Richards, J.A. *Remote Sensing Digital Image Analysis*; Springer: Berlin/Heidelberg, Germany, 2022; Volume 5.
34. Fan, X.; Hu, Z.; Zhao, Y.; Chen, J.; Wei, T.; Huang, Z. A small ship object detection method for satellite remote sensing data. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2024**, *17*, 11886–11898. [CrossRef]
35. Chaurasia, A.N.; Parmar, R.M.; Dave, M.G.; Krishnayya, N. Integrating field-and remote sensing data to perceive species heterogeneity across a climate gradient. *Sci. Rep.* **2024**, *14*, 42. [CrossRef]
36. Cavender-Bares, J.; Schneider, F.D.; Santos, M.J.; Armstrong, A.; Carnaval, A.; Dahlin, K.M.; Fatoyinbo, L.; Hurtt, G.C.; Schimel, D.; Townsend, P.A.; et al. Integrating remote sensing with ecology and evolution to advance biodiversity conservation. *Nat. Ecol. Evol.* **2022**, *6*, 506–519. [CrossRef] [PubMed]
37. Li, Y.; Fu, Y.; Lang, Z.; Cai, F. A High-Frequency and Real-Time Ground Remote Sensing System for Obtaining Water Quality Based on a Micro Hyper-Spectrometer. *Sensors* **2024**, *24*, 1833. [CrossRef]
38. Li, F.; Liu, Z.; Shen, W.; Wang, Y.; Wang, Y.; Ge, C.; Sun, F.; Lan, P. A remote sensing and airborne edge-computing based detection system for pine wilt disease. *IEEE Access* **2021**, *9*, 66346–66360. [CrossRef]
39. Singh, R.; Sharma, R. Streaming of High-Velocity Information using Dynamic Spatio-Temporal Query Processing. In Proceedings of the 2022 Second International Conference on Advanced Technologies in Intelligent Control, Environment, Computing & Communication Engineering (ICATIECE), Bangalore, India, 16–17 December 2022; IEEE: Piscataway, NJ, USA, 2022; pp. 1–6.
40. Rane, J.; Kaya, O.; Mallick, S.; Rane, N. Artificial intelligence-powered spatial analysis and ChatGPT-driven interpretation of remote sensing and GIS data. In *Generative Artificial Intelligence in Agriculture, Education, and Business*; Deep Science Publishing: London, UK, 2024; pp. 162–217.
41. Hong, D.; Li, C.; Zhang, B.; Yokoya, N.; Benediktsson, J.; Chanussot, J. Multimodal artificial intelligence foundation models: Unleashing the power of remote sensing big data in earth observation. *Innovation* **2024**, *2*, 100055. [CrossRef]
42. Wu, Z.; Sun, J.; Zhang, Y.; Wei, Z.; Chanussot, J. Recent developments in parallel and distributed computing for remotely sensed big data processing. *Proc. IEEE* **2021**, *109*, 1282–1305. [CrossRef]
43. Das, H.; Barik, R.K.; Dubey, H.; Roy, D.S. Cloud computing for geospatial big data analytics. In *Intelligent Edge, Fog and Mist Computing*; Springer: Berlin/Heidelberg, Germany, 2019.
44. Sun, J.; Zhang, Y.; Wu, Z.; Zhu, Y.; Yin, X.; Ding, Z.; Wei, Z.; Plaza, J.; Plaza, A. An efficient and scalable framework for processing remotely sensed big data in cloud computing environments. *IEEE Trans. Geosci. Remote Sens.* **2019**, *57*, 4294–4308. [CrossRef]
45. Luo, Y.; Wu, C.; Xue, Y.; Cai, G.; Hu, Y.; Wang, Z. A remote sensing application workflow and its implementation in remote sensing service grid node. In *Advances in Computational Methods in Sciences and Engineering 2005 (2 vols)*; CRC Press: Boca Raton, FL, USA, 2022; pp. 353–360.
46. Caon, M.; Ros, P.M.; Martina, M.; Bianchi, T.; Magli, E.; Membibre, F.; Ramos, A.; Latorre, A.; Kerr, M.; Wiehle, S.; et al. Very low latency architecture for earth observation satellite onboard data handling, compression, and encryption. In Proceedings of the 2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS, Brussels, Belgium, 11–16 July 2021; IEEE: Piscataway, NJ, USA, 2021; pp. 7791–7794.
47. Yu, H.; Hua, Z.; Miao, X.; Wang, S.; Pan, G.; An, J.; Svensson, T.; Thompson, J.; Yanikomeroglu, H. Computer Vision-Based Joint Space Sensing and Communication Systems: Non-Source, Autonomy, and Low Latency. *IEEE Wirel. Commun.* **2023**, *30*, 60–68. [CrossRef]

48. Sohl, M.A.; Mahmood, S.A. Low-Cost UAV in Photogrammetric Engineering and Remote Sensing: Georeferencing, DEM Accuracy, and Geospatial Analysis. *J. Geovis. Spat. Anal.* **2024**, *8*, 14. [CrossRef]
49. Wang, T.; Liang, Y.; Shen, X.; Zheng, X.; Mahmood, A.; Sheng, Q.Z. Edge computing and sensor-cloud: Overview, solutions, and directions. *ACM Comput. Surv.* **2023**, *55*, 1–37. [CrossRef]
50. Deng, X.; Liu, P.; Liu, X.; Wang, R.; Zhang, Y.; He, J.; Yao, Y. Geospatial big data: New paradigm of remote sensing applications. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2019**, *12*, 3841–3851. [CrossRef]
51. Sawadogo, P.; Darmont, J. On data lake architectures and metadata management. *J. Intell. Inf. Syst.* **2021**, *56*, 97–120. [CrossRef]
52. Xu, C.; Du, X.; Yan, Z.; Fan, X. ScienceEarth: A big data platform for remote sensing data processing. *Remote Sens.* **2020**, *12*, 607. [CrossRef]
53. Zhang, D.; Ren, L.; Shafiq, M.; Gu, Z. A lightweight privacy-preserving system for the security of remote sensing images on iot. *Remote Sens.* **2022**, *14*, 6371. [CrossRef]
54. Le Vine, D.M.; Dinnat, E.P. Sensitivity of wide bandwidth radiometer for remote sensing of ocean salinity. *IEEE Trans. Geosci. Remote Sens.* **2021**, *60*, 5301517.
55. Jiang, J.; Johansen, K.; Tu, Y.H.; McCabe, M.F. Multi-sensor and multi-platform consistency and interoperability between UAV, Planet CubeSat, Sentinel-2, and Landsat reflectance data. *GISci. Remote Sens.* **2022**, *59*, 936–958. [CrossRef]
56. Maniadaki, M.; Papathanasopoulos, A.; Mitrou, L.; Maria, E.A. Reconciling remote sensing technologies with personal data and privacy protection in the European Union: Recent developments in Greek legislation and application perspectives in environmental law. *Laws* **2021**, *10*, 33. [CrossRef]
57. Anderson, M.C.; Yang, Y.; Xue, J.; Knipper, K.R.; Yang, Y.; Gao, F.; Hain, C.R.; Kustas, W.P.; Cawse-Nicholson, K.; Hulley, G.; et al. Interoperability of ECOSTRESS and Landsat for mapping evapotranspiration time series at sub-field scales. *Remote Sens. Environ.* **2021**, *252*, 112189. [CrossRef]
58. Khalsa, S.J.S.; Percivall, G. Standardization in Geoscience Remote Sensing. In Proceedings of the IGARSS 2022—2022 IEEE International Geoscience and Remote Sensing Symposium, Kuala Lumpur, Malaysia, 17–22 July 2022; IEEE: Piscataway, NJ, USA, 2022; pp. 4676–4678.
59. Percivall, G.; Simonis, I. Advancements in Open Geospatial Standards for Photogrammetry and Remote Sensing from Ogc. *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.* **2016**, *41*, 705–709. [CrossRef]
60. Lary, D.J.; Alavi, A.H.; Gandomi, A.H.; Walker, A.L. Machine learning in geosciences and remote sensing. *Geosci. Front.* **2016**, *7*, 3–10. [CrossRef]
61. Zhang, L.; Zhang, L. Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities. *IEEE Geosci. Remote Sens. Mag.* **2022**, *10*, 270–294. [CrossRef]
62. Yuan, Q.; Shen, H.; Li, T.; Li, Z.; Li, S.; Jiang, Y.; Xu, H.; Tan, W.; Yang, Q.; Wang, J.; et al. Deep learning in environmental remote sensing: Achievements and challenges. *Remote Sens. Environ.* **2020**, *241*, 111716. [CrossRef]
63. Camps-Valls, G.; Tuia, D.; Zhu, X.X.; Reichstein, M. *Deep Learning for the Earth Sciences: A Comprehensive Approach to Remote Sensing, Climate Science and Geosciences*; John Wiley & Sons: Hoboken, NJ, USA, 2021.
64. Wu, X.; Hong, D.; Chanussot, J. Convolutional neural networks for multimodal remote sensing data classification. *IEEE Trans. Geosci. Remote Sens.* **2021**, *60*, 5517010. [CrossRef]
65. Zheng, Z.; Wan, Y.; Zhang, Y.; Xiang, S.; Peng, D.; Zhang, B. CLNet: Cross-layer convolutional neural network for change detection in optical remote sensing imagery. *ISPRS J. Photogramm. Remote Sens.* **2021**, *175*, 247–267. [CrossRef]
66. Hang, L.; Cai, G. CNN based detection of building roofs from high resolution satellite images. *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.* **2020**, *42*, 187–192. [CrossRef]
67. Zhang, C.; Wei, S.; Ji, S.; Lu, M. Detecting large-scale urban land cover changes from very high resolution remote sensing images using CNN-based classification. *ISPRS Int. J. Geo-Inf.* **2019**, *8*, 189. [CrossRef]
68. Hong, D.; Gao, L.; Yokoya, N.; Yao, J.; Chanussot, J.; Du, Q.; Zhang, B. More diverse means better: Multimodal deep learning meets remote-sensing imagery classification. *IEEE Trans. Geosci. Remote Sens.* **2020**, *59*, 4340–4354. [CrossRef]
69. Petrovska, B.; Zdravevski, E.; Lameski, P.; Corizzo, R.; Štajduhar, I.; Lerga, J. Deep learning for feature extraction in remote sensing: A case-study of aerial scene classification. *Sensors* **2020**, *20*, 3906. [CrossRef] [PubMed]
70. Dong, Y.; Gao, D.; Li, Y.; Shi, G.; Liu, D. Degradation estimation recurrent neural network with local and non-local priors for compressive spectral imaging. *IEEE Trans. Geosci. Remote Sens.* **2024**, *62*, 5520115. [CrossRef]
71. Pérez-Suay, A.; Adsuara, J.E.; Piles, M.; Martínez-Ferrer, L.; Díaz, E.; Moreno-Martínez, A.; Camps-Valls, G. Interpretability of recurrent neural networks in remote sensing. In Proceedings of the IGARSS 2020—2020 IEEE International Geoscience and Remote Sensing Symposium, Waikoloa, HI, USA, 26 September–2 October 2020; IEEE: Piscataway, NJ, USA, 2020; pp. 3991–3994.
72. Patra, R.K.; Patil, S.N.; Falkowski-Gilski, P.; Łubniewski, Z.; Poongodan, R. Feature Weighted Attention—Bidirectional Long Short Term Memory Model for Change Detection in Remote Sensing Images. *Remote Sens.* **2022**, *14*, 5402. [CrossRef]
73. Su, H.; Zhang, T.; Lin, M.; Lu, W.; Yan, X.H. Predicting subsurface thermohaline structure from remote sensing data based on long short-term memory neural networks. *Remote Sens. Environ.* **2021**, *260*, 112465. [CrossRef]

74. Zaheer, M.; Ahmed, A.; Smola, A.J. Latent LSTM allocation: Joint clustering and non-linear dynamic modeling of sequence data. In Proceedings of the International Conference on Machine Learning, PMLR, Sydney, Australia, 6–11 August 2017; pp. 3967–3976.
75. Ma, Y.; Chen, S.; Ermon, S.; Lobell, D.B. Transfer learning in environmental remote sensing. *Remote Sens. Environ.* **2024**, *301*, 113924. [CrossRef]
76. Cui, B.; Chen, X.; Lu, Y. Semantic segmentation of remote sensing images using transfer learning and deep convolutional neural network with dense connection. *IEEE Access* **2020**, *8*, 116744–116755. [CrossRef]
77. Thirumaladevi, S.; Swamy, K.V.; Sailaja, M. Remote sensing image scene classification by transfer learning to augment the accuracy. *Meas. Sens.* **2023**, *25*, 100645. [CrossRef]
78. Zhang, M.; Li, W.; Tao, R.; Wang, S. Transfer learning for optical and SAR data correspondence identification with limited training labels. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2020**, *14*, 1545–1557. [CrossRef]
79. Nowakowski, A.; Del Rosso, M.P.; Zachar, P.; Spiller, D.; Gabara, G.; Barretta, D.; Kalinowska, K.B.; Choromański, K.; Wilkowski, A.; Sebastianelli, A.; et al. Transfer Learning in Earth Observation Data Analysis: A review. *IEEE Geosci. Remote Sens. Mag.* **2024**, *early access*.
80. Kakogeorgiou, I.; Karantzalos, K. Evaluating explainable artificial intelligence methods for multi-label deep learning classification tasks in remote sensing. *Int. J. Appl. Earth Obs. Geoinf.* **2021**, *103*, 102520. [CrossRef]
81. Höhl, A.; Obadic, I.; Fernández-Torres, M.A.; Oliveira, D.; Zhu, X.X. Recent Trends Challenges and Limitations of Explainable AI in Remote Sensing. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, Seattle, WA, USA, 17–18 June 2024; pp. 8199–8205.
82. Schiff, D.; Rakova, B.; Ayesh, A.; Fanti, A.; Lennon, M. Principles to practices for responsible AI: Closing the gap. *arXiv* **2020**, arXiv:2006.04707.
83. Huang, F.; Jiang, S.; Li, L.; Zhang, Y.; Zhang, Y.; Zhang, R.; Li, Q.; Li, D.; Shangguan, W.; Dai, Y. Applications of Explainable artificial intelligence in Earth system science. *arXiv* **2024**, arXiv:2406.11882.
84. Marmanis, D.; Datcu, M.; Esch, T.; Stilla, U. Deep learning earth observation classification using ImageNet pretrained networks. *IEEE Geosci. Remote Sens. Lett.* **2015**, *13*, 105–109. [CrossRef]
85. Zhang, X.; Han, L.; Han, L.; Zhu, L. How well do deep learning-based methods for land cover classification and object detection perform on high resolution remote sensing imagery? *Remote Sens.* **2020**, *12*, 417. [CrossRef]
86. Persello, C.; Wegner, J.D.; Hänsch, R.; Tuia, D.; Ghamisi, P.; Koeva, M.; Camps-Valls, G. Deep learning and earth observation to support the sustainable development goals: Current approaches, open challenges, and future opportunities. *IEEE Geosci. Remote Sens. Mag.* **2022**, *10*, 172–200. [CrossRef]
87. Schumann, G.; Giustarini, L.; Tarpanelli, A.; Jarihani, B.; Martinis, S. Flood modeling and prediction using earth observation data. *Surv. Geophys.* **2023**, *44*, 1553–1578. [CrossRef]
88. Naushad, R.; Kaur, T.; Ghaderpour, E. Deep transfer learning for land use and land cover classification: A comparative study. *Sensors* **2021**, *21*, 8083. [CrossRef]
89. Selvam, K.P.; Ramos-Pollan, R.; Kalaitzis, F. Rapid Adaptation of Earth Observation Foundation Models for Segmentation. *arXiv* **2024**, arXiv:2409.09907.
90. Taskin, G.; Aptoula, E.; Ertürk, A. Explainable AI for Earth observation: Current methods, open challenges, and opportunities. In *Advances in Machine Learning and Image Analysis for GeoAI*; Elsevier: Amsterdam, The Netherlands, 2024; pp. 115–152.
91. Tuia, D.; Schindler, K.; Demir, B.; Zhu, X.X.; Kochupillai, M.; Džeroski, S.; van Rijn, J.N.; Hoos, H.H.; Del Frate, F.; Datcu, M.; et al. Artificial Intelligence to Advance Earth Observation: A review of models, recent trends, and pathways forward. *IEEE Geosci. Remote Sens. Mag.* **2024**, *early access*.
92. Mai, G.; Huang, W.; Sun, J.; Song, S.; Mishra, D.; Liu, N.; Gao, S.; Liu, T.; Cong, G.; Hu, Y.; et al. On the opportunities and challenges of foundation models for geospatial artificial intelligence. *arXiv* **2023**, arXiv:2304.06798.
93. Dritsas, E.; Trigka, M. Federated Learning for IoT: A Survey of Techniques, Challenges, and Applications. *J. Sens. Actuator Netw.* **2025**, *14*, 9. [CrossRef]
94. Ivić, M. Artificial intelligence and geospatial analysis in disaster management. *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.* **2019**, *42*, 161–166. [CrossRef]
95. Linardos, V.; Drakaki, M.; Tzionas, P.; Karnavas, Y.L. Machine learning in disaster management: Recent developments in methods and applications. *Mach. Learn. Knowl. Extr.* **2022**, *4*, 446–473. [CrossRef]
96. Sharifi, A.; Mahdipour, H. Utilizing geospatial artificial intelligence for remote sensing applications. *Environ. Earth Sci.* **2024**, *83*, 274. [CrossRef]
97. Liangpei, Z.; Jiang, H.; Qianqian, Y.; Yi, X.; Qiangqiang, Y. Data-driven multi-source remote sensing data fusion: Progress and challenges. *Acta Geod. Et Cartogr. Sin.* **2022**, *51*, 1317.
98. Gao, Y.; Zhang, M.; Wang, J.; Li, W. Cross-scale mixing attention for multisource remote sensing data fusion and classification. *IEEE Trans. Geosci. Remote Sens.* **2023**, *61*, 5507815. [CrossRef]

99. Judah, A.; Hu, B. The integration of multi-source remotely-sensed data in support of the classification of wetlands. *Remote Sens.* **2019**, *11*, 1537. [CrossRef]
100. Liu, X.; Zheng, C.; Wang, G.; Zhao, F.; Tian, Y.; Li, H. Integrating Multi-Source Remote Sensing Data for Forest Fire Risk Assessment. *Forests* **2024**, *15*, 2028. [CrossRef]
101. Li, J.; Zhang, J.; Yang, C.; Liu, H.; Zhao, Y.; Ye, Y. Comparative analysis of pixel-level fusion algorithms and a new high-resolution dataset for SAR and optical image fusion. *Remote Sens.* **2023**, *15*, 5514. [CrossRef]
102. Chen, C.; He, X.; Guo, B.; Zhao, X.; Chu, Y. A pixel-level fusion method for multi-source optical remote sensing image combining the principal component analysis and curvelet transform. *Earth Sci. Inform.* **2020**, *13*, 1005–1013.
103. Yang, X.X. *Urban Remote Sensing: Monitoring, Synthesis and Modeling in the Urban Environment*; John Wiley & Sons: Hoboken, NJ, USA, 2021.
104. Magee, A. *A Decision-Level Multi-Sensor Data Fusion Approach to Land Cover Classification*; National University of Ireland: Maynooth, Ireland, 2023.
105. Barik, R.K.; Misra, C.; Lenka, R.K.; Dubey, H.; Mankodiya, K. Hybrid mist-cloud systems for large scale geospatial big data analytics and processing: Opportunities and challenges. *Arab. J. Geosci.* **2019**, *12*, 32. [CrossRef]
106. Li, W.; Awais, M.; Ru, W.; Shi, W.; Ajmal, M.; Uddin, S.; Liu, C. Review of Sensor Network-Based Irrigation Systems Using IoT and Remote Sensing. *Adv. Meteorol.* **2020**, *2020*, 8396164. [CrossRef]
107. Sishodia, R.P.; Ray, R.L.; Singh, S.K. Applications of remote sensing in precision agriculture: A review. *Remote Sens.* **2020**, *12*, 3136. [CrossRef]
108. Fadda, M.; Anedda, M.; Girau, R.; Pau, G.; Giusto, D.D. A social internet of things smart city solution for traffic and pollution monitoring in cagliari. *IEEE Internet Things J.* **2022**, *10*, 2373–2390. [CrossRef]
109. Blasch, E.; Pham, T.; Chong, C.Y.; Koch, W.; Leung, H.; Braines, D.; Abdelzaher, T. Machine learning/artificial intelligence for sensor data fusion–opportunities and challenges. *IEEE Aerosp. Electron. Syst. Mag.* **2021**, *36*, 80–93. [CrossRef]
110. Kush, J.C. Integrating Sensor Technologies with Conversational AI: Enhancing Context-Sensitive Interaction Through Real-Time Data Fusion. *Sensors* **2025**, *25*, 249. [CrossRef]
111. Putrama, I.M.; Martinek, P. Heterogeneous data integration: Challenges and opportunities. *Data Brief* **2024**, *56*, 110853. [CrossRef] [PubMed]
112. Pondi, B.; Appel, M.; Pebesma, E. OpenEOcubes: An open-source and lightweight R-based RESTful web service for analyzing earth observation data cubes. *Earth Sci. Inform.* **2024**, *17*, 1809–1818. [CrossRef]
113. Munawar, H.S.; Qayyum, S.; Ullah, F.; Sepasgozar, S. Big data and its applications in smart real estate and the disaster management life cycle: A systematic analysis. *Big Data Cogn. Comput.* **2020**, *4*, 4. [CrossRef]
114. Karagiannopoulou, A.; Tsertou, A.; Tsimiklis, G.; Amditis, A. Data fusion in earth observation and the role of citizen as a sensor: A scoping review of applications, methods and future trends. *Remote Sens.* **2022**, *14*, 1263. [CrossRef]
115. Patanè, G.; Spagnuolo, M. *Heterogeneous Spatial Data: Fusion, Modeling, and Analysis for GIS Applications*; Springer Nature: Berlin/Heidelberg, Germany, 2022.
116. Liu, B.; Li, Q.; Zheng, Z.; Huang, Y.; Deng, S.; Huang, Q.; Liu, W. A Review of Multi-Source Data Fusion and Analysis Algorithms in Smart City Construction: Facilitating Real Estate Management and Urban Optimization. *Algorithms* **2025**, *18*, 30. [CrossRef]
117. Usmani, M. Fusion of Remote Sensing and Citizen Science Information Through Machine Learning for Geospatial Analysis. Ph.D Thesis, University of Trento, Trento, Italy, 2024.
118. Sun, W.; Shan, J.; Wang, Z.; Wang, L.; Lu, D.; Jin, Z.; Yu, K. Geospatial analysis of urban expansion using remote sensing methods and data: A case study of Yangtze River Delta, China. *Complexity* **2020**, *2020*, 3239471. [CrossRef]
119. Yussupov, A.; Suleimenova, R.Z. Use of Remote Sensing Data for Environmental Monitoring of Desertification. 2023. Available online: https://www.tj.kyushu-u.ac.jp/evergreen/contents/EG2023-10_1_content/pdf/p300-307.pdf (accessed on 15 December 2024).
120. Horak, J.; Voumik, L.C.; Popescu, G.H. Remote sensing data fusion techniques, multimodal behavioral predictive and mobile location analytics, and spatial cognition and context awareness algorithms in the metaverse economy. *Linguist. Philos. Investig.* **2023**, *22*, 77–93.
121. Kaku, K. Satellite remote sensing for disaster management support: A holistic and staged approach based on case studies in Sentinel Asia. *Int. J. Disaster Risk Reduct.* **2019**, *33*, 417–432. [CrossRef]
122. Gupta, R.; Shah, M. Rescuenet: Joint building segmentation and damage assessment from satellite imagery. In Proceedings of the 2020 25th International Conference on Pattern Recognition (ICPR), Milan, Italy, 10–15 January 2021; IEEE: Piscataway, NJ, USA, 2021; pp. 4405–4411.
123. Al Shafian, S.; Hu, D. Integrating Machine Learning and Remote Sensing in Disaster Management: A Decadal Review of Post-Disaster Building Damage Assessment. *Buildings* **2024**, *14*, 2344. [CrossRef]

124. Zheng, Z.; Zhong, Y.; Wang, J.; Ma, A.; Zhang, L. Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters. *Remote Sens. Environ.* **2021**, *265*, 112636. [CrossRef]
125. Wellmann, T.; Lausch, A.; Andersson, E.; Knapp, S.; Cortinovis, C.; Jache, J.; Scheuer, S.; Kremer, P.; Mascarenhas, A.; Kraemer, R.; et al. Remote sensing in urban planning: Contributions towards ecologically sound policies? *Landsc. Urban Plan.* **2020**, *204*, 103921. [CrossRef]
126. Alahi, M.E.E.; Sukkuea, A.; Tina, F.W.; Nag, A.; Kurdthongmee, W.; Suwannarat, K.; Mukhopadhyay, S.C. Integration of IoT-enabled technologies and artificial intelligence (AI) for smart city scenario: Recent advancements and future trends. *Sensors* **2023**, *23*, 5206. [CrossRef] [PubMed]
127. Haas, J. Remote Sensing of Urbanization and Environmental Impacts. Ph.D. Thesis, KTH Royal Institute of Technology, Stockholm, Sweden, 2016.
128. Arifin, S.; Yudhatama, D. Analysis of land use spatial pattern change of town development using remote sensing. *Int. J. Remote Sens. Earth Sci.* **2018**, *15*, 93–102. [CrossRef]
129. Ouchra, H.; Belangour, A.; Erraissi, A. A comprehensive study of using remote sensing and geographical information systems for urban planning. *Internetworking Indones. J.* **2022**, *14*, 15–20.
130. Dritsas, E.; Trigka, M. Machine Learning for Blockchain and IoT Systems in Smart Cities: A Survey. *Future Internet* **2024**, *16*, 324. [CrossRef]
131. Segarra, J.; Buchaillot, M.L.; Araus, J.L.; Kefauver, S.C. Remote sensing for precision agriculture: Sentinel-2 improved features and applications. *Agronomy* **2020**, *10*, 641. [CrossRef]
132. Lu, B.; Dao, P.D.; Liu, J.; He, Y.; Shang, J. Recent advances of hyperspectral imaging technology and applications in agriculture. *Remote Sens.* **2020**, *12*, 2659. [CrossRef]
133. Jeong, S.; Ko, J.; Kim, M.; Kim, J. Construction of an unmanned aerial vehicle remote sensing system for crop monitoring. *J. Appl. Remote Sens.* **2016**, *10*, 026027. [CrossRef]
134. Bharadiya, J.P.; Tzenios, N.T.; Reddy, M. Forecasting of crop yield using remote sensing data, agrarian factors and machine learning approaches. *J. Eng. Res. Rep.* **2023**, *24*, 29–44. [CrossRef]
135. Martos, V.; Ahmad, A.; Cartujo, P.; Ordoñez, J. Ensuring agricultural sustainability through remote sensing in the era of agriculture 5.0. *Appl. Sci.* **2021**, *11*, 5911. [CrossRef]
136. Yang, J.; Gong, P.; Fu, R.; Zhang, M.; Chen, J.; Liang, S.; Xu, B.; Shi, J.; Dickinson, R. The role of satellite remote sensing in climate change studies. *Nat. Clim. Change* **2013**, *3*, 875–883. [CrossRef]
137. Purkis, S.J.; Klemas, V.V. *Remote Sensing and Global Environmental Change*; John Wiley & Sons: Hoboken, NJ, USA, 2011.
138. Rhee, J.; Im, J. Meteorological drought forecasting for ungauged areas based on machine learning: Using long-range climate forecast and remote sensing data. *Agric. For. Meteorol.* **2017**, *237*, 105–122. [CrossRef]
139. Florath, J.; Keller, S.; Abarca-del Rio, R.; Hinz, S.; Staub, G.; Weinmann, M. Glacier monitoring based on multi-spectral and multi-temporal satellite data: A case study for classification with respect to different snow and ice types. *Remote Sens.* **2022**, *14*, 845. [CrossRef]
140. Filchev, L.; Chanev, M. Remote Sensing in Climate Change Research. In Proceedings of the The International Conference on Environmental Protection and Disaster Risks, Sofia, Bulgaria, 4–6 June 2024; Springer: Berlin/Heidelberg, Germany, 2024; pp. 147–166.
141. Dube, T.; Shekede, M.D.; Massari, C. Remote sensing for water resources and environmental management. *Remote Sens.* **2022**, *15*, 18. [CrossRef]
142. Kseňak, L.; Pukanská, K.; Bartoš, K.; Blišt'an, P. Assessment of the usability of SAR and optical satellite data for monitoring spatio-temporal changes in surface water: Bodrog river case study. *Water* **2022**, *14*, 299. [CrossRef]
143. Adjovu, G.E.; Stephen, H.; James, D.; Ahmad, S. Overview of the application of remote sensing in effective monitoring of water quality parameters. *Remote Sens.* **2023**, *15*, 1938. [CrossRef]
144. Adikari, K.E.; Shrestha, S.; Ratnayake, D.T.; Budhathoki, A.; Mohanasundaram, S.; Dailey, M.N. Evaluation of artificial intelligence models for flood and drought forecasting in arid and tropical regions. *Environ. Model. Softw.* **2021**, *144*, 105136. [CrossRef]
145. Oerke, E.C. Remote sensing of diseases. *Annu. Rev. Phytopathol.* **2020**, *58*, 225–252. [CrossRef] [PubMed]
146. Bhatti, U.A.; Tang, H.; Wu, G.; Marjan, S.; Hussain, A. Deep learning with graph convolutional networks: An overview and latest applications in computational intelligence. *Int. J. Intell. Syst.* **2023**, *2023*, 8342104. [CrossRef]
147. Rodó, X.; Ballester, J.; Curcoll, R.; Boyard-Micheau, J.; Borràs, S.; Morguí, J.A. Revisiting the role of environmental and climate factors on the epidemiology of Kawasaki disease. *Ann. N. Y. Acad. Sci.* **2016**, *1382*, 84–98. [CrossRef] [PubMed]
148. Bhunia, G.S.; Shit, P.K. *Geospatial Analysis of Public Health*; Springer: Berlin/Heidelberg, Germany, 2019.
149. Musse, M.A.; Barona, D.A.; Rodriguez, L.M.S. Urban environmental quality assessment using remote sensing and census data. *Int. J. Appl. Earth Obs. Geoinf.* **2018**, *71*, 95–108. [CrossRef]
150. Lee, J.G.; Kang, M. Geospatial big data: Challenges and opportunities. *Big Data Res.* **2015**, *2*, 74–81. [CrossRef]

151. Pan, S.; Li, Y.; Xu, Z.; Chong, Y. Distributed storage algorithm for geospatial image data based on data access patterns. *PLoS ONE* **2015**, *10*, e0133029. [CrossRef]
152. Kritikos, K.; Rousakis, Y.; Kotzinos, D. A cloud-based, geospatial linked data management system. In *Transactions on Large-Scale Data-and Knowledge-Centered Systems XX: Special Issue on Advanced Techniques for Big Data Management*; Springer: Berlin/Heidelberg, Germany, 2015; pp. 59–89.
153. Butenko, O.; Horelik, S.; Zynyuk, O. Geospatial data processing characteristics for environmental monitoring tasks. *Archit. Civ. Eng. Environ.* **2020**, *13*, 103–114. [CrossRef]
154. Rathore, M.M.U.; Paul, A.; Ahmad, A.; Chen, B.W.; Huang, B.; Ji, W. Real-time big data analytical architecture for remote sensing application. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2015**, *8*, 4610–4621. [CrossRef]
155. Gao, F.; Yue, P.; Cao, Z.; Zhao, S.; Shangguan, B.; Jiang, L.; Hu, L.; Fang, Z.; Liang, Z. A multi-source spatio-temporal data cube for large-scale geospatial analysis. *Int. J. Geogr. Inf. Sci.* **2022**, *36*, 1853–1884. [CrossRef]
156. Nistor, A. Development of Geospatial Technologies by Using Artificial Intelligence. *FAIMA Bus. Manag. J.* **2024**, *12*, 61–68.
157. Yin, J.; Qi, C.; Chen, Q.; Qu, J. Spatial-spectral network for hyperspectral image classification: A 3-D CNN and Bi-LSTM framework. *Remote Sens.* **2021**, *13*, 2353. [CrossRef]
158. Yin, L.; Xie, J. Multi-temporal-spatial-scale temporal convolution network for short-term load forecasting of power systems. *Appl. Energy* **2021**, *283*, 116328. [CrossRef]
159. Veeramachaneni, V. Edge Computing: Architecture, Applications, and Future Challenges in a Decentralized Era. *Recent Trends Comput. Graph. Multimed. Technol.* **2025**, *7*, 8–23.
160. Saleem, U.; Liu, Y.; Jangsher, S.; Tao, X.; Li, Y. Latency minimization for D2D-enabled partial computation offloading in mobile edge computing. *IEEE Trans. Veh. Technol.* **2020**, *69*, 4472–4486. [CrossRef]
161. Samikwa, E.; Voigt, T.; Eriksson, J. Flood prediction using IoT and artificial neural networks with edge computing. In Proceedings of the 2020 International Conferences on Internet of Things (iThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCom) and IEEE Smart Data (SmartData) and IEEE Congress on Cybermatics (Cybermatics), Rhodes Island, Greece, 2–6 November 2020; IEEE: Piscataway, NJ, USA, 2020; pp. 234–240.
162. Peyman, M.; Copado, P.J.; Tordecilla, R.D.; Martins, L.d.C.; Xhafa, F.; Juan, A.A. Edge computing and IoT analytics for agile optimization in intelligent transportation systems. *Energies* **2021**, *14*, 6309. [CrossRef]
163. Hu, X.; Wen, H. Research on model compression for embedded platform through quantization and pruning. *J. Phys. Conf. Ser.* **2021**, *2078*, 012047. [CrossRef]
164. Beltrán, E.T.M.; Pérez, M.Q.; Sánchez, P.M.S.; Bernal, S.L.; Bovet, G.; Pérez, M.G.; Pérez, G.M.; Celdrán, A.H. Decentralized federated learning: Fundamentals, state of the art, frameworks, trends, and challenges. *IEEE Commun. Surv. Tutor.* **2023**, *25*, 2983–3013. [CrossRef]
165. Cowls, J.; Tsamados, A.; Taddeo, M.; Floridi, L. The AI gambit: Leveraging artificial intelligence to combat climate change—opportunities, challenges, and recommendations. *AI Soc.* **2023**, *38*, 283–307. [CrossRef] [PubMed]
166. Giuliani, G.; Masó, J.; Mazzetti, P.; Nativi, S.; Zabala, A. Paving the way to increased interoperability of earth observations data cubes. *Data* **2019**, *4*, 113. [CrossRef]
167. Kochupillai, M.; Kahl, M.; Schmitt, M.; Taubenböck, H.; Zhu, X.X. Earth observation and artificial intelligence: Understanding emerging ethical issues and opportunities. *IEEE Geosci. Remote Sens. Mag.* **2022**, *10*, 90–124. [CrossRef]
168. Li, J.; Li, Y.; He, L.; Chen, J.; Plaza, A. Spatio-temporal fusion for remote sensing data: An overview and new benchmark. *Sci. China Inf. Sci.* **2020**, *63*, 140301. [CrossRef]
169. Henderson, M.; Gallina, J.; Brett, M. Methods for accelerating geospatial data processing using quantum computers. *Quantum Mach. Intell.* **2021**, *3*, 4. [CrossRef]
170. Mahmud, N.; Haase-Divine, B.; MacGillivray, A.; El-Araby, E. Quantum dimension reduction for pattern recognition in high-resolution spatio-spectral data. *IEEE Trans. Comput.* **2020**, *71*, 1–12. [CrossRef]
171. Makarov, A.; Taddei, M.M.; Osaba, E.; Franceschetto, G.; Villar-Rodriguez, E.; Oregi, I. Optimization of Image Acquisition for Earth Observation Satellites via Quantum Computing. In Proceedings of the International Conference on Intelligent Data Engineering and Automated Learning, Évora, Portugal, 22–24 November 2023; Springer: Berlin/Heidelberg, Germany, 2023; pp. 3–14.
172. Vista, F.; Iacovelli, G.; Grieco, L.A. Hybrid quantum-classical scheduling optimization in UAV-enabled IoT networks. *Quantum Inf. Process.* **2023**, *22*, 47. [CrossRef]
173. Sarin, S.; Singh, S.K.; Kumar, S.; Goyal, S. Geospectra: Leveraging quantum-SAR and deep learning for enhanced geolocation in urban environments. *J. Supercomput.* **2025**, *81*, 223. [CrossRef]
174. Duraimurugan, N.; Vaseekaran, K.; Vikram, V. Quantum Foliage Finder Revolutionizing Environmental Monitoring and Conservation With Quantum Computing Through Deep Learning. In *Real-World Applications of Quantum Computers and Machine Intelligence*; IGI Global Scientific Publishing: Hershey, PA, USA, 2025; pp. 263–276.

175. Sebastianelli, A.; Zaidenberg, D.A.; Spiller, D.; Le Saux, B.; Ullo, S.L. On circuit-based hybrid quantum neural networks for remote sensing imagery classification. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2021**, *15*, 565–580. [CrossRef]
176. Qin, D.; Chen, Y.; Li, Y. Error statistics and scalability of quantum error mitigation formulas. *Npj Quantum Inf.* **2023**, *9*, 35. [CrossRef]
177. Saxena, A.K. Advancing Location Privacy in Urban Networks: A Hybrid Approach Leveraging Federated Learning and Geospatial Semantics. *Int. J. Inf. Cybersecur.* **2023**, *7*, 58–72.
178. Ahmed, L.; Ahmad, K.; Said, N.; Qolomany, B.; Qadir, J.; Al-Fuqaha, A. Active learning based federated learning for waste and natural disaster image classification. *IEEE Access* **2020**, *8*, 208518–208531. [CrossRef]
179. Pandya, S.; Srivastava, G.; Jhaveri, R.; Babu, M.R.; Bhattacharya, S.; Maddikunta, P.K.R.; Mastorakis, S.; Piran, M.J.; Gadekallu, T.R. Federated learning for smart cities: A comprehensive survey. *Sustain. Energy Technol. Assess.* **2023**, *55*, 102987. [CrossRef]
180. Algarni, F.; Khan, M.A.; Alawad, W.; Halima, N.B. P3S: Pertinent privacy-preserving scheme for remotely sensed environmental data in smart cities. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2023**, *16*, 5905–5918. [CrossRef]
181. Moreno-Álvarez, S.; Paoletti, M.E.; Sanchez-Fernandez, A.J.; Rico-Gallego, J.A.; Han, L.; Haut, J.M. Federated learning meets remote sensing. *Expert Syst. Appl.* **2024**, *255*, 124583. [CrossRef]
182. Ray, P.P. Federated Learning for Intelligent IoT Systems: Background, Frameworks, and Optimization Techniques. In *Model Optimization Methods for Efficient and Edge AI: Federated Learning Architectures, Frameworks and Applications*; John Wiley & Sons: Hoboken, NJ, USA, 2025; pp. 241–280.
183. Saha, R.; Misra, S.; Deb, P.K. FogFL: Fog-assisted federated learning for resource-constrained IoT devices. *IEEE Internet Things J.* **2020**, *8*, 8456–8463. [CrossRef]
184. Elkordy, A.R.; Avestimehr, A.S. HeteroSAg: Secure aggregation with heterogeneous quantization in federated learning. *IEEE Trans. Commun.* **2022**, *70*, 2372–2386. [CrossRef]
185. Abdulmuttalib, H.; Al Doori, M.; Barsi, Á.; Blaschke, T.; Gao, Y.; Kugler, Z.; Lang, S.; Szabo, G.; Tiede, D. Remote Sensing Data Quality in the Era of AI. *ISPRS Ann. Photogramm. Remote Sens. Spat. Inf. Sci.* **2024**, *10*, 1–11. [CrossRef]
186. Liu, R.; Mi, L.; Chen, Z. AFNet: Adaptive fusion network for remote sensing image semantic segmentation. *IEEE Trans. Geosci. Remote Sens.* **2020**, *59*, 7871–7886. [CrossRef]
187. Mohammadpour, P.; Viegas, C. Applications of Multi-Source and Multi-Sensor Data Fusion of Remote Sensing for Forest Species Mapping. In *Advances in Remote Sensing for Forest Monitoring*; John Wiley & Sons: Hoboken, NJ, USA, 2022; pp. 255–287.
188. Jony, R.I. Multimodal Data Fusion of Remote Sensing and Social Media Using Machine Learning for Natural Disaster Detection and Assessment. Ph.D. Thesis, Queensland University of Technology, Brisbane, Australia, 2023.
189. Sabri, S.; Rajabifard, A.; Chen, Y.; Chen, N.; Sheng, H. Geospatial Understanding of Sustainable Urban Analytics Using Remote Sensing. *Remote Sens.* **2022**, *14*, 2748. [CrossRef]
190. Cao, X.; Madria, S. Efficient geospatial data collection in iot networks for mobile edge computing. In Proceedings of the 2019 IEEE 18th International Symposium on Network Computing and Applications (NCA), Cambridge, MA USA, 26–28 September 2019; IEEE: Piscataway, NJ, USA, 2019; pp. 1–10.
191. Bajaj, K.; Sharma, B.; Singh, R. Implementation analysis of IoT-based offloading frameworks on cloud/edge computing for sensor generated big data. *Complex Intell. Syst.* **2022**, *8*, 3641–3658. [CrossRef]
192. Hernández Olcina, J.; Anquela Julián, A.B.; Martín Furones, Á.E. Navigating latency hurdles: An in-depth examination of a cloud-powered GNSS real-time positioning application on mobile devices. *Sci. Rep.* **2024**, *14*, 14668. [CrossRef] [PubMed]
193. Xing, J.; Sieber, R. Challenges of Using XAI for Geographic Data Analytics. In Proceedings of the 1st International Workshop on Methods, Models, and Resources for Geospatial Knowledge Graphs and GeoA, Poznań, Poland, 27 September 2021; Newcastle University: Newcastle upon Tyne, UK, 2021.
194. Thunki, P.; Reddy, S.R.B.; Raparthi, M.; Maruthi, S.; Dodda, S.B.; Ravichandran, P. Explainable AI in Data Science-Enhancing Model Interpretability and Transparency. *Afr. J. Artif. Intell. Sustain. Dev.* **2021**, *1*, 1–8.
195. Masrur, A.; Yu, M.; Mitra, P.; Peuquet, D.; Taylor, A. Interpretable machine learning for analysing heterogeneous drivers of geographic events in space-time. *Int. J. Geogr. Inf. Sci.* **2022**, *36*, 692–719. [CrossRef]
196. Quarati, A.; De Martino, M.; Rosim, S. Geospatial open data usage and metadata quality. *ISPRS Int. J. Geo-Inf.* **2021**, *10*, 30. [CrossRef]
197. Dziob, D.; Krupiński, M.; Woźniak, E.; Gabryszewski, R. Interdisciplinary teaching using satellite images as a way to introduce remote sensing in secondary school. *Remote Sens.* **2020**, *12*, 2868. [CrossRef]
198. Hojati, M. Distributed Spatial Data Sharing: A New Era in Sharing Spatial Data. Ph.D. Thesis, Wilfrid Laurier University, Waterloo, ON, Canada, 2023.